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On a Pickup to Delivery Drone Routing Problem: Models and algorithms

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ABSTRACT

A new variant of the Pickup and Delivery Routing problem is presented. Given a set of customers, facilities, a depot, and a homogeneous fleet of drones, the Pickup to Delivery Drone Routing Problem (PDDRP) aims to find a drone scheduling such that a drone serves the customer's order from a set of available facilities. Each drone starts in the depot, flies to pickup the customer's order in a facility, and continues its flight to deliver the parcel to a customer. Then, the drone begins another service, and once its last service is completed, it returns to the depot. The objective is to minimize the makespan associated with the drone fleet. The layer of facilities forcing drones to visit one of them to pickup the parcel makes the problem different from traditional pickup and delivery routing problems. Three mixed-linear programming models are presented to obtain optimal solutions for the problem. The first model is related to the multiple Traveling Salesman Problem (m-TSP), the second is associated with the Parallel Machine Scheduling Problem (PMS), and the third was developed specifically for the new problem.

Given the high computational complexity of the PDDRP, a Greedy Randomized Adaptive Search Procedure (GRASP) was designed to find near-optimal solutions when exact approaches cannot achieve (near) optimal solutions. Computational experiments show that a commercial solver could solve only small problem instances. GRASP can find reasonable solutions in a short time when medium and large instance sizes need to be solved. Finally, is shown that some routing problems for delivery, allowing truck-drone collaboration, could be formulated as an extension of PMS.

1. Introduction

1.1. Context and motivation

The Vehicle Routing Problem (VRP) has been well studied in the literature; see Braekers et al. (2016), Golden et al. (2008), Toth and Vigo (2014) and Vidal et al. (2020), where problems, models, algorithms, and applications are presented. The richness of this class of discrete optimization is awesome. In the last decades, due to e-commerce, mainly the operation of last-mile delivery has forced the study of different logistic problems. A central problem in this context is the planning of vehicle routing for delivery, assuming that the problem of depot location and the size of the vehicle fleet were already solved. Thus, depending on several factors like the location and number of customers, the type of vehicles conforming to the vehicle fleet, and the type of operations to be executed by the vehicles, the application of VRP models in the real world is fundamental. Unmanned Aerial Vehicles (UAVs), more known as Drones, have started to participate in parcel delivery in the last few years. The use of drones has greatly developed in the last years. Their use in Agriculture, Surveillance, and Delivery have been well known in the literature. Several surveys have made evident the rapid evolution of the literature about it. Some of they are, Chung et al. (2020), Khoufi et al. (2019), Macrina et al. (2020), Moshref-Javadi and Winkenbach (2021), Otto et al. (2018), Rojas-Viloria et al. (2021) and Shakhatreh et al. (2019). Particularly, the study of delivery by drones and trucks in tandem has been intensively developed since Murray and Chu (2015). Models, algorithms, and applications have been studied specifying different variants in routing problems using a truck-drone system, and it is still a hot area. The study of routing problems using only a fleet of drones has had an increasing interest in the scientific community, principally related to some applications, for example, in Healthcare (Ghelichi et al., 2021; Hii et al., 2019; Jiang et al., 2021; Sedov et al., 2020) and Humanitarian Logistics (Akram et al., 2020; Ejaz et al., 2020; Holzmann et al., 2021).

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Several reasons exist for the success of the use of drones. A drone is a cheap, light, and fast vehicle, and its development is in continuous evolution. According to the rapid growth of the technological characteristics of drones, several companies in the logistic area have been conducting significant innovations. Wing, a Google parent company, has been working on pilot tests for its drones in Virginia (x company, 2022). With its drone Parcelcopter, since 2013, DHL has been testing new improvements. In 2016, the company concluded a three-month test of its third parcelcopter generation in Bavaria under challenging climate conditions in the Alps (dpdhl, 2022). Walmart requested permission from the Federal Aviation Administration (FAA) to test its delivery drones for commercial purposes (The wired shopper, 2021). In 2019, UPS was granted FAA approval for retail drone deliveries (The verge, 2019). The company announced that it is expanding its services for healthcare customers and preparing new models of drones. Zipline International, an American start-up, has been working since 2016 in healthcare applications by using drones in Rwanda and Ghana (Adu-Gyamfi et al., 2021). A recent practical implementation of a drone for fast delivery was presented by flytrex (2022). Additional details about drone innovations in companies can be found in Macrina et al. (2020).

Despite the significant development of drones in the last years, some drawbacks for delivery operations exist; drones cannot operate under demanding climate conditions, and the battery's duration limits them. However, the technological development of drones allows an improved performance every year. Next, the problem to be studied and the contributions are defined.

1.2. The problem and the contribution

In the last years, the study of parcel delivery using the collaboration of trucks and drones has attracted the interest of many researchers. This research line has been surveyed by Chung et al. (2020), Moshref-Javadi and Winkenbach (2021) and Rojas-Viloria et al. (2021). There is literature studying problems, models, and algorithms for drone delivery when a fleet of drones serves a set of customers. The well-known Pickup and Delivery Problem (PDP) is related to the problem here defined. This problem has extensively been referenced in the literature, see Battarra et al. (2014), Berbeglia et al. (2007) and Savelsbergh and Sol (1995). In the PDP, two types of customers are defined, one part of them asks for a pickup operation, and the other asks for a delivery operation. Also, the two types of customers have to be visited exactly once. Our problem defines a set of customers to which one delivery is asked. Also are defined a set of facilities (or suppliers) from which every customer could be served. Thus, a drone should visit a facility to pickup a parcel to be delivered to a customer. This operation justifies our proposal to name the problem Pickup to Delivery. Additionally, a facility could be visited several times by a drone to serve a different customer each time. Therefore, the basic ideas behind these two problems are very different.

From the point of view of PDP applications, an important case is when the pick operation corresponds to reverse logistics. The vehicle picks up a parcel at a customer's location and then delivers it to the depot. In our problem it is different, only delivery operations are possible on the customers. Furthermore, in the specific literature about the collaboration of trucks and drones, most existing papers also adhere to the characteristics of PDP. We give a detailed definition of our problem and present an example illustrating the particularities of the problem below.

In the new problem, the Pickup to Delivery Drone Routing Problem (PDDRP), it is assumed that, in a region, there is a system composed of a set of customers each one with a order of products, a set of facilities supplying a set of products, and a homogeneous fleet of drones initially located in a depot. The system's task is to analyze the customer's orders and defines a schedule of the drones in such a way each customer's order is served by a drone. A customer orders products to be consolidated in only one facility (we assume that each facility can serve all the products ordered by the customers), completing a parcel to be delivered at the customer's home. The route of each drone starts in the depot, travels to a facility where it picks up a parcel, and then travels to deliver in a customer location. Then, from the customer's location just served, the drone either travels to a facility to begin a new service or returns to the depot finishing its operation. It is also assumed that the operation of the fleet of drones could take several hours, so it is necessary to have time to make drone maintenance in the depot. Then, the fleet of drones starts a new round of delivery. Potential applications of this problem are the supply of medical kits in the context of urban neighborhoods, or even in humanitarian logistics where the medicines could be delivered in several zones where roads were destroyed by a disaster.

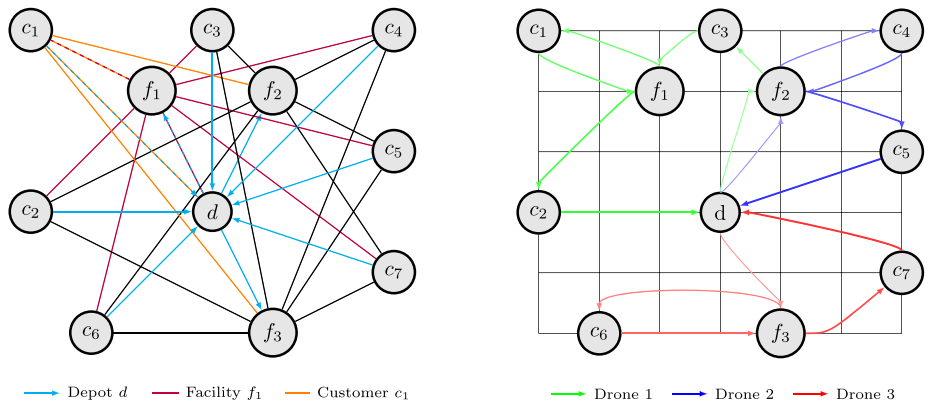
Several objective functions could be considered in the problem, for example, the makespan, the total time spent by the fleet of drones, and some function of costs associated with the use of drones. In this paper is used the makespan.

Fig. 1(a) illustrates a network with a depot (node d), three facilities (nodes f_1, f_2 and f_3) and seven customers ($c_1, \ldots, c_7$). The size of the drone fleet is three. The cyan colored arcs are connected with d. Thus, these arcs take direction from the depot to each facility, and from each customer to the depot. To illustrate the connections of facilities, there are purple colored arcs from f_1 to every customer, which are bi-directed, and one from d to f_1. Similarly, the orange colored arcs denotes the bi-directed connections of c_1 with each facility, and that directed from c_1 to d. The black colored arcs are the remaining connections. Fig. 1(b) shows a feasible solution for the problem where the route of each drone is colored. The route of the drone 1 is colored green, and it is: $d - f_2 - c_3 - f_1 - c_1 - f_1 - c_2 - d$, the route of drone 2 is colored blue, and it is: $d - f_2 - c_4 - f_2 - c_5 - d$, and the route of the drone 3 is colored red, and it is: $d - f_3 - c_6 - f_3 - c_7 - d$. In this paper, we assumed that drones charge and swap their batteries in the facilities and that this operation takes a short constant time. Therefore, it is not considered in the modeling of this problem.

The article contributes to the literature as follows:

- We define a new drone routing problem for parcel delivery (PDDRP). Three mathematical programming formulations are presented for the problem. We created a set of problem instances and conducted computational experiments to test the performance of the formulations. We used a CPLEX solver to solve the optimization models.
- Since PDDRP is an NP-Hard problem, obtaining optimal solutions is possible from solving the optimization models, only for a limited size of the test instances. Therefore, we design a GRASP metaheuristic to find solutions for instances that are difficult or impossible to solve by the exact algorithms. The contribution of the exact approaches and the metaheuristic GRASP was analyzed, highlighting the relevance of the two algorithmic parts of GRASP.
- Mixed Integer Lineal Programming (MILP) formulations allow getting lower bounds from the feasible solutions obtained. These bounds, together with the value of the best feasible solutions given by exact approaches and GRASP, were used to get the GAP of all the test instances.
- We extend the formulation of a known parallel scheduling machine problem to a new variant where two types of a homogeneous fleet of vehicles exist. It is shown that the Parallel Drone Scheduling Traveling Salesman Problem with Drop-Pickup (Ham, 2018), could be formulated using the extended model. The extension opens the study for other applications of truck-drone routing problems.

The paper continues with the analysis of related work in Section 2. In Section 3, the characteristics of the problem are provided, and three MILP formulations are detailed. Section 4 describes in detail the GRASP variants proposed. In Section 5, the results of the algorithms implemented are illustrated and analyzed. The extension of the parallel scheduling machine formulation is presented in Section 6. Finally, Section 7 presents conclusions and future research.



(a) PDDRP instance with depot d, three facilities f, and seven customers c

(b) Feasible solution for the instance when three drones are used

Fig. 1. One PDDRP instance and one of its feasible solution for three drones.

2. Literature review

The literature about logistic problems aided by drones has grown a lot lately. Several review papers have surveyed a variety of optimization problems, considering mathematical programming formulations, solution algorithms, and applications, see Chung et al. (2020), Khoufi et al. (2019), Otto et al. (2018), Rojas-Viloria et al. (2021), and Shakhatreh et al. (2019). The use of drones in the context of the last mile delivery has been reviewed in Boysen et al. (2021). Most emphasis has received the called truck-drone routing problem. This problem includes several variants considering different protocols of collaboration between fleets of trucks and drones. Therefore, we start the review with this class of routing problems.

Ham (2018) consider a PDP including several depots, a fleet of trucks, and a fleet of drones. Drop and pickup drone operations are allowed in such a way that after completing a drop in a customer, the drone could return to a depot or directly fly to another customer for pickup. The author proposed modeling the problem as an unrelated parallel machine scheduling with sequence-dependent setup, precedence-relationship, and reentrant. He proposed a constraint programming approach to solve the problem that was tested with problem instances of m-truck, m-drone, m-depot, and hundred-customer distributed across an 8-mile square region. Therefore, an optimization model was not proposed for the problem in this paper. In Karak and Abdelghany (2019) is a one-to-many-to-one PDP, as the vehicle and the drones are dispatched from one depot to deliver commodities to customers, and pickup commodities from the customers and transport them back to the depot. They also formulated the problem as a mixed-integer program, which minimizes the vehicle and drone routing cost to serve all customers. Additionally, they proposed an extension of the classic Clarke and Wright algorithm and conducted experiments to analyze the performance of the developed algorithms. Yu et al. (2021) extended the van-based robot pickup and delivery research by integrating multiple pickup and delivery modes in one trip to increase the efficiency of distribution systems. In the collaboration van-robot, parking nodes execute operations drop and pickup. These nodes can be visited multiple times by a van-robot. A two-echelon optimization formulation was constructed for the problem, and, additionally, an adaptive large neighborhood search was proposed. The algorithms were tested by using a set of instances. A case study considering realistic scenarios was also analyzed.

Since our problem considers only a fleet of drones, the literature review continues focused mainly on this class of delivery problems. Boysen et al. (2021) discuss several logistic problems related to delivery with a fleet of drones. One of these problems is the setup of infrastructure, where the optimal location of depots and the location planning problem for drone recharging stations are studied. Another problem is the staffing and fleet sizing, where optimization models are proposed, for example, for the integrated problem of depot location and drone fleet sizing. Also, they analyze some drone routing problems using only a fleet of drones for delivery. We comment about it below.

To situate the structure of the Drone Routing Problem (DRP), we take the point of view from Boysen et al. (2021) where the base DRP is a delivery problem containing one depot, a set of customers, and a fleet of drones. Note that the problem is trivial when only one drone is used. The base DRP is a particular case of the PMS. Drones assume the role of machines and customers the role of jobs. Therefore, the literature on DRP typically considers extensions of the base DRP. One of the first papers extending DRP is Dorling et al. (2017). This paper assumes that there is one depot, a set of customers, and a fleet of drones. It is considered drones can transport several parcels simultaneously in a sortie, visiting thus several customers. They also assumed that a set of charged batteries was available in the depot before the operation. So, when it is necessary, a drone trip to the depot to swap its battery. They define two delivery drone problems, the first consists of optimizing the number of drones in a fleet and the routes they fly. The second is the minimum cost drone delivery problem. In both problems, they consider constraints on the capacity of drones, reusability, and cost batteries. They formulate two similar MILP problems for the two variants of the problem and only change the objective function. For the first problem, they minimize the overall delivery time. Concerning the second problem, they minimize the cost of drones and energy. They also implemented a simulated annealing algorithm for solving large-size instances. Computational experiments with the model included the effect of drone reused and the optimizing battery weight.

Choi and Schonfeld (2017) studied the DRP of optimizing the drone fleet size for a service area by minimizing the total costs of the delivery system. This work assumes that a drone can lift multiple packages within its maximum payload and serve recipients in a service area of a given radius. They consider battery capacities to relate parcel payloads and flight ranges. San et al. (2016) studied the planning of a fleet of drones for customer delivery to minimize the completion

time. They implemented a genetic algorithm to solve DRP. There are no mathematical programming formulations and considerations about battery recharging in this paper. Troudi et al. (2018) studied the DRP of dimensioning from a forecast of deliveries of an urban perimeter, the size of the drone fleet, the stock of battery, and the strategy of battery charging. They proposed a routing optimization model for drone fleet sizing. The model's objective function simultaneously minimizes the total distance resulting from served parcels, the total used drones, and the total batteries used during these missions. Coelho et al. (2017) proposed a multiobjective optimization formulation to DRP with recharging stations in the context of smart cities. Their model evaluates the energy needed to recharge at the charging stations. Furthermore, they develop a matheuristic to solve large instances of the problem. Liu (2019) presented a DRP for the online fleet dispatch operations by drones. He assumed that a parcel might contain one or several types of meals and cold drinks. He formulated a MILP formulation that includes the particularities of the online fleet dispatch by drones. He also presented a case study to validate the proposed approach.

Finally, we analyze two papers with a relatively close relationship with our paper. They are studied in the context of classic PDPs. In Shi et al. (2009), it is considered a PDP variant where the commodity required by delivery customers is the same as that provided by pickup customers, which means that the commodity collected from a pickup customer can be directly supplied to a delivery customer. So, pickup customers in that paper correspond to the facilities in the present work. However, in that paper, pickup customers must be visited only once; in this work, a facility could not be visited or visited several times. Also, in the former, the objective function minimizes total travel distance, while, in the latter, the makespan is minimized.

In Al Chami et al. (2018), they study a variant of PDP, the Selective PDP with time windows and paired demands. In this variant, it is assumed that a request consists of a transportation load from pickup locations (suppliers) to delivery ones (customers). The selective aspect establishes that not all the customers must be visited. The paired demand means each supplier is associated with a specific customer. In that problem, it is made explicit that each node is visited at most once. Each vehicle describes a TSP route in a feasible solution, and then a feasible solution corresponds to an m-TSP solution. In this paper, a feasible solution is a more general route that allows visiting supplier nodes several times (see the feasible solution in Fig. 1(b) of the instance in Fig. 1(a)).

The following section presents notation, basic definitions, assumptions, and three mathematical formulations for the PDDRP.

3. Formulations for PDDRP

Firstly, we define the PDDRP and pose its computational complexity. Then, three MILP formulations are proposed. Notation, definitions, and fundamental assumptions are now presented.

- A1. In a region, there is a drone depot d, a set of facilities F, a set of customers C, a homogeneous fleet of drones V, and customer's orders.
- A2. Each customer has a demand for several products, which must be satisfied by a single drone, a single facility, and just one delivery. Also, the capacity of the drone could limit this demand. Therefore, we assume that the customer's order can be consolidated into one parcel in one facility and supported by the drone's capacity.
- A3. Each facility has a product inventory that must fully satisfy the demand of all customers for each product.
- A4. A drone can deliver a parcel to only one customer in a sortie.
- A5. After executing a delivery, the drone travels to a facility for a new service or the depot.
- A6. There is no additional distance traveled by a drone in the drone service when the customer receives the parcel.
- A7. Each drone starts its operation with a zero distance traveled.
- A8. Some drone travels are prohibited. For example, to go directly from the depot to a customer or from a facility to the depot.
- A9. When the drone is launched or retrieved at a particular node, set-up and recovery times are negligible. It is assumed that these times are short compared to the drone travel time and will not affect how the model determines the solution route. This assumption is made in several papers, for example, Kitjacharoenchai et al. (2019).

3.1. Network definitions

C is a set of customers, F is a set of facilities, and V is a set of drones. For convenience, $\{0\}$ is the depot. Also, $N = \{0\} \cup F \cup C$, $N_0 = \{0\} \cup C$, $\hat{N} = \{0\} \cup C \cup \{|C|+1\}$, and $N_+ = C \cup \{|C|+1\}$. $G = (N, A)$, where N is the set of nodes. $A = \{(0, f),\ f \in F\} \cup \{(f, c),\ f \in F,\ c \in C\} \cup \{(c, f),\ c \in C,\ f \in F\} \cup \{(c, 0),\ c \in C\}$ is the set of arcs. Each arc $(i, j) \in A$ has a non-negative distance t_{ij}.

Since we define three optimization formulations for the problem, it is necessary to establish the corresponding network components to every formulation.

- First Formulation. $\bar{G} = (N_0, \bar{A})$, where N_0 is the set of nodes, $\bar{A} = \{(i.j), i \in N_0, j \in N_0, i \neq j\}$ is the set of arcs. Every arc $(i, j) \in \bar{A}$ has a travel distance $\bar{t}_{ij}$.
- Second Formulation. $\hat{G} = (\hat{N}, \hat{A})$, where $\hat{N}$ is the set of nodes, $\hat{A} = \{(i, j),\ i \in N_0,\ j \in N_+, i \neq j\}$ is the set of arcs. Every arc $(i, j) \in \hat{A}$ has a travel distance $\hat{t}_{ij}$.
- Third Formulation. $G = (N, A)$, where N is the set of nodes, $A = \{(0, f),\ f \in F\} \cup \{(f, c),\ f \in F,\ c \in C\} \cup \{(c, f),\ c \in C,\ f \in F\} \cup \{(c, 0),\ c \in C\}$ is the set of arcs.

The travel distances $\hat{t}_{ij}$ and $\bar{t}_{ij}$ are obtained in function of the travel distances t_{ij}. Fig. 2 shows an example of the configuration of the travels in both networks $\hat{G}$ and $\bar{G}$. Figs. 2(a) and 2(b) show that, for a travel initiated in the depot and finalized in the customer c_5, or initiated in a customer c_2 and finalized in the customer c_5, respectively, there is at least one facility that induces a minimum travel distance of $\hat{t}_{ij}$ and $\bar{t}_{ij}$. Further, Fig. 2(c) shows that in the final trip of a drone, it travels directly from a customer to the depot.

Formally, these relations are defined as follows:

$$\bar{t}_{ij} = \hat{t}_{ij} = min\{t_{if} + t_{fj} :\ f \in F\}, i \in \{0\} \cup C,\ j \in C, i \neq j \tag{1}$$

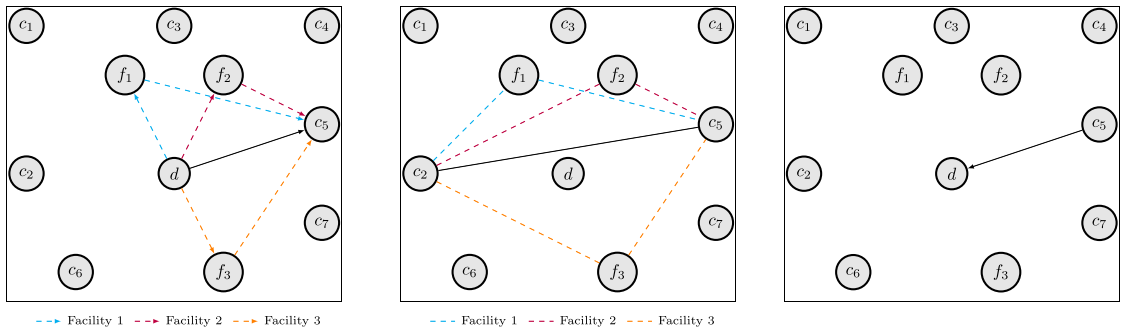
$$\bar{t}_{i0} = \hat{t}_{i,|C|+1} = t_{i0}, i \in C \tag{2}$$

PDDRP consists of finding an assignment of drones to customers such that all the customers are served while minimizing the value of the makespan w. In this problem, the makespan is the maximum distance traveled by a drone.

Lemma 1. *PDDRP is NP-Hard.*

Proof. It is easy to note that TSP is a special case of PDDRP. Let us consider a TSP instance I with set of nodes N, set of arcs (i, j) in A, and travel distances t_{ij}. Then, it is defined a PDDRP instance I', where there is a depot d, $d \in N$, a set of customers $N - \{d\}$, a set of $|N| - 1$ facilities f, each of them located in the same place of the customers, one drone located in the node d, and travel distances $t_{ij}, (i, j) \in N - \{d\}$. Then, an optimal solution for the PDDRP instance I' represents an optimal solution for the I instance of TSP.

Lemma 2. *PDDRP is trivial to solve if the number of drones is equal o greater than the number of customers.*



(a) travel from the depot to a customer (b) travel between customers (c) travel from a customer to the depot

Fig. 2. Example to illustrate travel distances $\hat{t}_{ij}$ and $\bar{t}_{ij}$.

3.2. First Formulation: Modeling PDDRP as an extension of m-TSP

In m-TSP, it is aimed to find one route for each one of m salesmen, who all start and end at the depot, such that each customer is visited exactly once in the salesman's route, and the total cost of visiting all customers is minimized. Bektas (2006) presented an excellent overview of the problem discussing mathematical formulations, solution procedures and practical applications. He adapted a three-index flow VRP formulation from Christofides et al. (1981) to a minisum m-TSP model, excluding the original vehicle capacity constraints. In this model, the objective function aims to minimize the total travel costs of the operations. Thus, we replaced this objective function with a makespan. Also, we added constraints ensuring that the value of the makespan is greater than the distance traveled by each drone. The mathematical formulation is given as follows.

Decision variables
$\bar{x}_{ijv} \in \{0,1\}$, $\bar{x}_{ijv} = 1$ if drone v travels from the node i to the node j, $\bar{x}_{ijv} = 0$ otherwise, where $(i,j) \in \bar{A}$, $v \in V$.
w: the value of makespan.
u_i: auxiliary variables for sub-tours elimination, where $i \in N_0$.

$$\min w \tag{3}$$

subject to

$$\sum_{v\in V}\sum_{(i,j)\in\bar{A}} \bar{x}_{ijv} = 1, j \in C \tag{4}$$

$$\sum_{(0,j)\in\bar{A}} \bar{x}_{0jv} = 1, v \in V \tag{5}$$

$$\sum_{(i,h)\in\bar{A}} \bar{x}_{ihv} - \sum_{(h,j)\in\bar{A}} \bar{x}_{hjv} = 0, h \in N_0, v \in V \tag{6}$$

$$u_i - u_j + |N_0| \sum_{v\in V} \bar{x}_{ijv} \le |N_0| - 1,\ (i,j) \in \bar{A}, i \neq 0 \tag{7}$$

$$u_0 = 0 \tag{8}$$

$$w \ge \sum_{(i,j)\in\bar{A}} \bar{t}_{ij}\bar{x}_{ijv},\ v \in V \tag{9}$$

$$\bar{x}_{ijv} \in \{0,1\},\ (i,j) \in \bar{A},\ v \in V \tag{10}$$

$$u_i \ge 0,\ i \in N_0 \tag{11}$$

$$w \ge 0 \tag{12}$$

The objective function (3) minimizes the makespan. Constraints (4) assure that each customer should be served exactly once. Constraints (5) restrict that all drones must be launched from the depot. Constraints (6) guarantee the flow conservation. Constraints (7) and (8) are about the subtour elimination. Constraints (9) ensure that w is greater than the total distance traveled by each drone. Constraints (10)–(12) correspond to the domain of the decision variables.

3.3. Second Formulation: Modeling PDDRP as an extension of Parallel Machines Scheduling

In Unlu and Mason (2010) several MILP formulations were evaluated for PMS problems. Specifically, a network-based version is selected. Using some assumptions, this model can solve TSP/VRP problems. Therefore, it can also be used to solve PDDRP. For this purpose, each job can be represented as a node of the network, and the completion time of jobs is the distance traveled on serving the customers and land on the depot. Additionally, it is necessary to incorporate a setup time between each pair of jobs as the travel distance among each pair of nodes and do not consider the release date and the processing time of jobs used in the original model. Several scheduling articles include setup times, see Balakrishnan et al. (1999), Lin and Hsieh (2014) and Zhu and Heady (2000). For convenience, an extra depot node for the return of the drones is used. Then, the modified model is formulated to replace the original objective function for a makespan expression.

Decision variables
$\hat{x}_{ijv} \in \{0,1\}$, $\hat{x}_{ijv} = 1$ if drone v travels from the node i to the node j, $\hat{x}_{ijv} = 0$ otherwise, where $(i,j) \in \hat{A}$, $v \in V$
w: the value of makespan
D_{iv}: distance used to reach the node i using drone v, where $i \in \hat{N}$, $v \in V$

$$\min w \tag{13}$$

subject to

$$\sum_{v\in V}\sum_{(i,j)\in\hat{A}} \hat{x}_{ijv} = 1,\ j \in C \tag{14}$$

$$\sum_{(0,j)\in\hat{A}} \hat{x}_{0jv} = 1, v \in V \tag{15}$$

$$\sum_{(i,h)\in\hat{A}} \hat{x}_{ihv} - \sum_{(h,j)\in\hat{A}} \hat{x}_{hjv} = 0, v \in V, h \in C \tag{16}$$

$$D_{jv} \ge D_{iv} - M + (\hat{t}_{ij} + M)\hat{x}_{ijv}, (i,j) \in \hat{A}, v \in V \tag{17}$$

$$D_{0v} = 0,\ v \in V \tag{18}$$

$$w \ge D_{iv},\ i \in N^+,\ v \in V \tag{19}$$

$$\hat{x}_{ijv} \in \{0,1\},\ (i,j) \in \hat{A},\ v \in V \tag{20}$$

$$D_{iv} \geq 0,\ i \in N^{+},\ v \in V \tag{21}$$

$$w \geq 0 \tag{22}$$

The objective function (13) minimizes the makespan. Constraints (14) guarantee that a single drone must serve every customer. Constraints (15) ensure that all drones are launched from the depot. Constraints (16) are the flow conservation on the customer nodes. Subtours are eliminated by constraints (17) and (18). These constraints also set the distance used to reach all customers. Constraints (19) restrict that w should not be less than the distance traveled by every drone. Constraints (20)–(22) define the domain of the decision variables.

3.4. Third Formulation: Modeling PDDRP using a direct approach

Both formulations above are extensions of known optimization problems. Now, we propose a new formulation that directly considers the conditions of the problem. In this model, the facilities F are considered explicitly in the network. These facilities can be visited more than once for a drone to generate some cycles. For example, in Fig. 1(b), the route for drone 2 is $d - f_2 - c_4 - f_2 - c_5 - d$. Therefore, this route contains a cycle $f_2 - c_4 - f_2$. The model needs an additional parameter p_{if} to manage the construction of feasible solutions. This parameter will be used in the subtour elimination constraints, and it is defined as follows: $p_{ij} = \begin{cases} -1, & i = 0 \\ 1, & i = f,\ \ i \in N, f \in F \\ 0, & otherwise \end{cases}$

Decision variables.

$x_{ijv} \in \{0,1\}$, where $x_{ijv} = 1$ if drone v travels from the node i to the node j, $x_{ijv} = 0$ otherwise, where $(i,j) \in A$, $v \in V$.

$y_{fv} \in \{0,1\}$, where $y_{fv} = 1$ if drone v travels for picking a product to the facility f, $y_{fv} = 0$ otherwise, where $f \in F$, $v \in V$.

$z_{ijfv} \in \{0,1\}$, where $z_{ijfv} = 1$ if in the travel of drone v, there is a path that join the depot with facility f using arc (i,j), $z_{ijfv} = 0$ otherwise, where $(i,j) \in A, f \in F, v \in V$

w: the value of the makespan.

The formulation of this model is now presented.

$$\min w \tag{23}$$

subject to

$$\sum_{(i,j)\in A} \sum_{v \in V} x_{ijv} = 1,\ j \in C \tag{24}$$

$$\sum_{(0,j)\in A} x_{0jv} = 1,\ v \in V \tag{25}$$

$$\sum_{(i,j)\in A} x_{ijv} - \sum_{(j,h)\in A} x_{jhv} = 0,\ j \in F \cup C,\ v \in V \tag{26}$$

$$w \geq \sum_{(i,j)\in A} t_{ij} x_{ijv},\ v \in V \tag{27}$$

$$x_{ijv} \leq y_{jv},\ v \in V,\ (i,j) \in A:\ j \in F \tag{28}$$

$$z_{ijfv} \leq x_{ijv},\ (i,j) \in A,\ f \in F,\ v \in V \tag{29}$$

$$\sum_{(i,h)\in A} z_{ihfv} - \sum_{(h,j)\in A} z_{hjfv} = p_{hf} y_{fv},\ f \in F,\ v \in V,\ h \in N \tag{30}$$

$$x_{ijv} \in \{0,1\},\ (i,j) \in A,\ v \in V \tag{31}$$

$$y_{fv} \in \{0,1\},\ f \in F,\ v \in V \tag{32}$$

$$z_{ijfv} \in \{0,1\},\ (i,j) \in A,\ f \in F,\ v \in V \tag{33}$$

$$w \geq 0 \tag{34}$$

Eq. (23) defines the minimization of the makespan as the objective function. Constraints (24) establish that each customer must be served by only one drone. Constraints (25) impose that every drone has to leave the depot. Constraints (26) show that if drone v delivers to customer j from a facility i, then it must leave from node j to the depot or a facility. Constraints (27) show that the value of the makespan is at least the maximum distance traveled by a drone. Constraints (28) establish that a drone v only can traveling from node i to node j if drone v visits node j. Constraints (29) shows that if the arc (i,j) is not visited by the drone v then drone v cannot visit facility f when travel from i to j. Constraints (30) are the subtours elimination, assuring that every drone travels a connected route. So (30) define a path that connects the depot with all the facilities visited by the drone. (31)–(34) correspond to the definition of the domain of the decision variables.

As we will show in Section 5, all the three models here are presented to find optimal solutions for a set of instances of small size. We refer to m-TSP, PMS, and Direct approach as First, Second, and Third Formulation. The following section presents a GRASP approach for finding near-optimal solutions for greater instances.

4. A GRASP metaheuristic for the problem

For small instances is assumed that near-optimal or optimal solution could be obtained by an exact approach in a reasonable time, according to the MILP formulations described in the past section. The complexity of PDDRP has been established as an NP-Hard problem. For this reason, a GRASP metaheuristic that allows solving medium and large-size instances is now presented.

GRASP is a well-known metaheuristic that has a good performance when applied to complex optimization problems (Resende & Ribeiro, 2014). In particular, recent papers proposed GRASP approaches to solving difficult truck-drone problems, see Gómez-Lagos et al. (2021) and Ha et al. (2018). GRASP is an iterative algorithm that includes two sequential procedures in each iteration. First, a constructive heuristic based on a greedy randomized strategy finds a feasible solution S followed by a local search procedure to improve S. This paper presents two variants of the greedy randomized algorithm and two variants of the subsequent local search procedure. In Algorithm 1 we present a pseudocode of the GRASP procedure.

Algorithm 1 Procedure GRASP(MaxIterations, seed)

```
f* ← ∞
for k = 1,.., MaxIterations do
    S ← GreedyRandomizedAlgorithm(Seed)
    S ← LocalSearch(S)
    if f(S) < f* then
        S* ← S
        f* ← f(S)
    end if
end for
return S*
```

4.1. Greedy Randomized procedures

Several variants of a Greedy Randomized procedure have been proposed in the literature. We use the well known value-based version (Resende & Ribeiro, 2014) and a sample greedy version (Resende & Werneck, 2004) to solve PDDRP.

The solution is made from constructing the drone routes that begin in the depot for both presented schemes. Thus, a candidate (a drone visiting a customer) is added to the partial solution using different

criteria in each step. Both procedures finalize when every customer is assigned to a drone route, and then each drone returns to the depot. The $random(a, b)$ function selects randomly a elements from the set b. The $update(I_{vc})$ function recalculate the values of the incremental costs I_{vc} for the remaining candidates CU. These functions are used in Algorithm 2 and Algorithm 3, explained below.

4.1.1. First Constructive: Value-based version

This constructive scheme is applied as given in Algorithm 2. It is based on the idea that the restricted candidate list (RCL) comprises the elements that induce the best (the smallest) incremental costs on the objective function at each step. With this aim, the parameter $\alpha \in [0, 1]$ determines the grade of greediness or randomness of the RCL construction. In a minimization problem, when $\alpha = 0$, it is said that the procedure is pure-greedy and when $\alpha = 1$ is pure-randomly.

Algorithm 2 First Constructive

Require: C, α, $\bar{t}$
  $S \leftarrow \emptyset$ ▷ Solution
  $S_v \leftarrow \{0\} : v \in V$ ▷ Route of drone v
  $I_{vc} \leftarrow \{\bar{t}_{0c}\} : v \in V, c \in C$ ▷ Incremental costs
  $CU \leftarrow C$ ▷ Customers not yet served
  while $|CU| > 0$ **do**
    $RCL \leftarrow \emptyset$ ▷ Restricted Candidate List
    $I_{min} \leftarrow min\{I_{vc}\} : v \in V$, $c \in CU$
    $I_{max} \leftarrow max\{I_{vc}\} : v \in V$, $c \in CU$
    for $v \in V$ **do**
      for $c \in CU$ **do**
        if $I_{vc} \leq I_{min} + \alpha(I_{max} - I_{min})$ **then**
          $RCL \leftarrow RCL \cup \{\{v, c\}\}$
        end if
      end for
    end for
    $elected \leftarrow random(1, RCL)$
    $S_{elected_1} \leftarrow S_{elected_1} \cup elected_2$
    $CU \leftarrow CU \backslash elected_2$
    $I_{vc} \leftarrow update(I_{vc})$
  end while
  for $v \in V$ **do**
    $S_v \leftarrow S_v \cup \{0\}$
  end for
  $S \leftarrow \{S_v\} \cup \{S_{v+1}\} : v \in V \backslash V_{|V|}$
return S

4.1.2. Second Constructive: Sample greedy version

The Second Constructive scheme is given in the Algorithm 3 and follows the idea that the list RCL is made up with p randomly selected items at each step. Then, a pure-greedy procedure is performed that aims to set the element that induces the smallest incremental cost on the objective function. Considering the multiple sizes of the studied networks, which implicates that the number of candidates at the first step is $|V| \times |C|$, the parameter $\%p \in [0, 1]$ standardizes this procedure. The $\%p = 0$ notation indicates that just one candidate will be included in the RCL, and $\%p = 1$ indicates that every candidate will be incorporated. Therefore, when $\%p = 1$, the procedure is pure-greedy and is pure-randomly when $\%p = 0$.

4.2. Local search procedures

Local search procedures allow improving the initial solution provided by the constructive phase. Two neighborhoods are implemented, and two variants that use different neighborhood structure combinations are considered. The used neighborhoods are Insertion Neighborhood and 2-opt Neighborhood. We present both of them below.

Algorithm 3 Second Constructive

Require: C, $\%p$, I_{vc}, $\bar{t}$
  $S \leftarrow \emptyset$ ▷ Solution
  $S_v \leftarrow \{0\} : v \in V$ ▷ Route of drone v
  $RCL \leftarrow \emptyset$ ▷ Restricted Candidate List
  $CU \leftarrow C$ ▷ Customers not yet served
  $I_{vc} \leftarrow \{\bar{t}_{0c}\} : v \in V, c \in C$ ▷ Incremental costs
  $p \leftarrow \%p \times |V| \times |C|$
  while $|CU| > 0$ **do**
    $q \leftarrow |V| \times |CU|$
    $p \leftarrow min\{p, q\}$
    $RCL \leftarrow random(p, I_{vc})$
    $elected \leftarrow argmin(RCL)$
    $S_{elected_1} \leftarrow S_{elected_1} \cup elected_2$
    $CU \leftarrow CU \backslash elected_2$
    $I_{vc} \leftarrow update(I_{vc})$
  end while
  for $v \in V$ **do**
    $S_v \leftarrow S_v \cup \{0\}$
  end for
  $S \leftarrow \{S_v\} \cup \{S_{v+1}\} : v \in V \backslash V_{|V|}$
return S

4.2.1. Neighborhood structures

The Insertion neighborhood (Soylu, 2015) can be applied for min-sum or makespan objectives. Each movement takes one customer of the longest drone route(s) in the makespan version and relocates it to another. One of these movements is illustrated in Fig. 3. In Fig. 3(a) the route of drone 2 is the longest and is formed by $d - c_2 - c_3 - c_5 - d$. Then, in Fig. 3(b) c_2 is inserted on the drone 1 route, between nodes d and c_1. Note that every solution has $|C_L| \times (|C| - |C_L| + |V_U| - 1)$ neighbors, where $|C_L|$ is the number of customers served on the longest route and $|V_U|$ is the number of used drones.

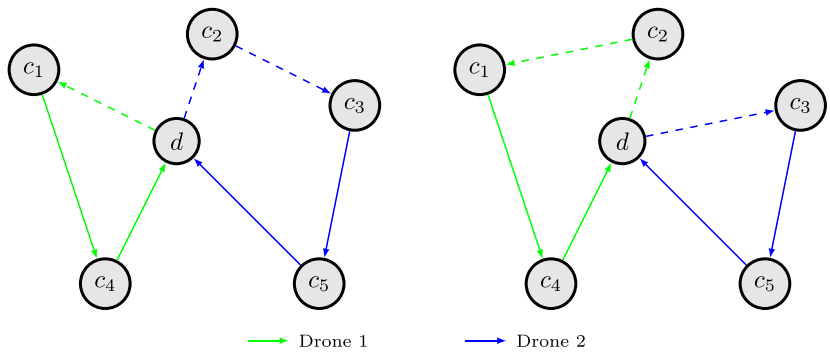
The 2-opt neighborhood structure was initially proposed for the TSP by Croes (1958). In this structure, the neighbors are characterized by an intra-route movement that removes two non-adjacent arcs and adds two new arcs. Since the improvements are made for each route separately, this neighborhood does not directly improve the makespan objective but can refine the Insertion Search.

An example is shown in Fig. 4, where the route of drone 2 is intervened. Here, the arcs $d - c_2$ and $c_3 - c_5$ (Fig. 4(a)) change order to form $d - c_3$ and $c_2 - c_5$ arcs (Fig. 4(b)). Since the route of the drones are directed, the change in the direction of the intermediate arcs ($c_2 - c_3$ to $c_3 - c_2$) should not affect on the arcs related to the depot node and unrelated to the movement ($c_5 - d$). On the other hand, the route of drone 1 does not have two non-adjacent arcs, so a 2-opt movement cannot be performed.

Concerning the neighborhood function, we proposed two local search procedures. In both versions, the methods were implemented with the best improvement strategy. Both local search procedures are described below.

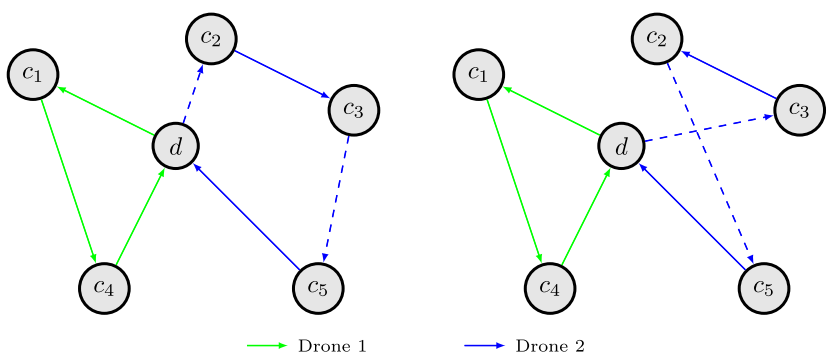
4.2.2. First Local Search procedure

The First Local Search procedure is presented in Algorithm 4. It uses only the Insertion Neighborhood. Every neighbor of the current solution S is explored, and those that induce the best improvement on the objective function are evaluated. Since the structure of the problem allows multiple solutions that satisfy this condition, S^{list} includes all of them. Then, the selection of the actual improvement is made considering the solution of S^{list} which increases as little as possible the distance of the route to which the insertion is made S'^{+}. The $dist(S'^{+})$ function calculates this distance. This sequence is running until finding a local optimum.



(a) A feasible solution (b) Neighbor of the feasible solution

Fig. 3. The Insertion neighborhood in the Local Search.



(a) A feasible solution (b) A neighbor of the feasible solution

Fig. 4. 2-opt neighborhood.

Algorithm 4 First Local Search

Require: S
  $improvement \leftarrow true$
  $S^{list} \leftarrow \emptyset$ ▷ List of best improvement solutions
  $r^{+} \leftarrow \emptyset$ ▷ List of route distances which include the inserted customer nodes
  while $improvement$ **do**
    $improvement \leftarrow false$
    for $S' \in IN(S)$ **do** ▷ Insertion Neighborhood
      if $f(S') < f(S)$ **then**
        $S_{min} \leftarrow S'$
        $improvement \leftarrow true$
      end if
    end for
    for $S' \in IN(S)$ **do**
      if $f(S') = f(S_{min})$ **then**
        $S^{list} \leftarrow S^{list} \cup S'$
        $r^{+} \leftarrow r^{+} \cup dist(S'^{+})$
      end if
    end for
    $S \leftarrow S^{list,i} \mid r_i^{+} = min\{r_i^{+}\} : i \in \{1, ..., |S^{list}|\}$
  end while
return S

4.2.3. Second Local Search procedure

The Second Local Search procedure is presented in Algorithm 5 and adds a phase of searching using a 2-opt neighborhood to the first strategy. Thus, an extensive search on the 2-opt neighborhood for improving every drone route is completed before an insertion move is made.

5. Computational experiments

This section presents the set of problem instances generated to study the algorithm's performance in Section 5.1. Then, the results of the experiments, along with an analysis of the algorithm's performance, are presented in Section 5.2. Finally, models and algorithms findings are presented in Section 5.3.

5.1. Data for experiments

To evaluate the performance of the optimization models and GRASP heuristics, we generate several sets of data divided into two main groups of instances, namely I_1 and I_2. We use the following terminology to denote an instance: $|C|c-|F|f-|V|v-I_\beta$, where β is the index of the main group. Specifically, we have been used $|C| \in \{20, 40, 60, 100\}$, $|F| \in \{3, 5, 10, 15\}$ and $|V| \in \{4, 6, 8\}$. Additionally, a special case is considered when $|C| = 150$, the amount of facilities varies as $|F| \in \{5, 10, 15, 20\}$, and $|V| \in \{4, 6, 8, 12\}$. Therefore, we generate a total of 128 instances. The methodology for the generation of the instances is accumulative. A bigger instance includes all the nodes of the smaller

Algorithm 5 Second Local Search

```
Require: S
  improvement_1 ← true
  S^list ← ∅                                  ▷ List of best improvement solutions
  r^+ ← ∅                ▷ List of route distances which include the inserted
  customer nodes
  while improvement_1 do
      improvement_1 ← false
      improvement_2 ← true
      while improvement_2 do
          improvement_2 ← false
          for S' ∈ 2optN(S) do                    ▷ 2-Opt Neighborhood
              if f(S') < f(S) then
                  S_min ← S'
                  improvement_2 ← true
              end if
          end for
          S ← S_min
      end while
      for S' ∈ IN(S) do                           ▷ Insertion Neighborhood
          if f(S') < f(S) then
              S_min ← S'
              improvement ← true
          end if
      end for
      for S' ∈ IN(S) do
          if f(S') = f(S_min) then
              S^list ← S^list ∪ S'
              r^+ ← r^+ ∪ dist(S'^+)
          end if
      end for
      S ← S^list,i | r_i^+ = min{r_i^+} : i ∈ {1, ..., |S^list|}
  end while
return S
```

instances, and new customers are added until the number of customers of the bigger instance is complete. The same criterion is applied to the construction of the facilities.

The coordinates of customer and facility nodes were randomly created within a 2-D region. The region is separated into three concentric squares whose dimensions are $width, length \in \{200, 600, 1000\}$ $units$. The distribution of nodes defines the customers concentrated in the periphery area and the facilities in the downtown area. To represent this network distribution, the percentage of customers $\%_c$ and facilities $\%_f$ to generate into the downtown, middle and periphery areas have been set as $\%_c \in \{0.1, 0.2, 0.7\}$ and $\%_f \in \{0.7, 0.2, 0.1\}$, respectively. The coordinates of the depot node are always (500, 500). Also, the minimum distance between each pair of nodes is 10 $units$. Fig. 5 shows three network configurations. Note that customers and facilities of $20c - 3f - I_1$ (Fig. 5(a)) are contained on $60c - 10f - I_1$ (Fig. 5(b)), and $150c - 20f - I_1$ (Fig. 5(c)) nodes. Also, $60c - 10f - I_1$ nodes are included into $150c - 20f - I_1$.

The process and raw data used to generate the test instances are more depth explained in Appendix and the processed data of the test instances are available via a web page. https://drive.google.com/drive/folders/1eqEH9JlsCB7em2zYLXtyKZVKvKSUZYc2

5.2. Results and analysis of the experiments

In this section, computational experiments are conducted to evaluate the performance of the implemented algorithms Section 5.2.1 presents and analyzes the results of the exact algorithms, while in Section 5.2.2, those of the metaheuristic versions.

5.2.1. Results of exact algorithms

Computational environment. CPLEX v. 12.9 was used to solve MILP problems, and the solver was executed on a Dell PowerEdge R/730 with Intel Xeon E5 2623-v3 3 GHz computer. We imposed an upper limit of 3600 s on each run of CPLEX.

Tables 1 and 2 show the values of the objective function, runtime and GAP found by CPLEX for each model, and for I_1 and I_2 instances, respectively. Let $UB(I)$ and $LB(I)$ be the values of the upper and lower bounds found by CPLEX for the instance I. Then, the GAPs shown in the tables are calculated as follows:

$$GAP(I) = \frac{UB(I) - LB(I)}{UB(I)} \tag{35}$$

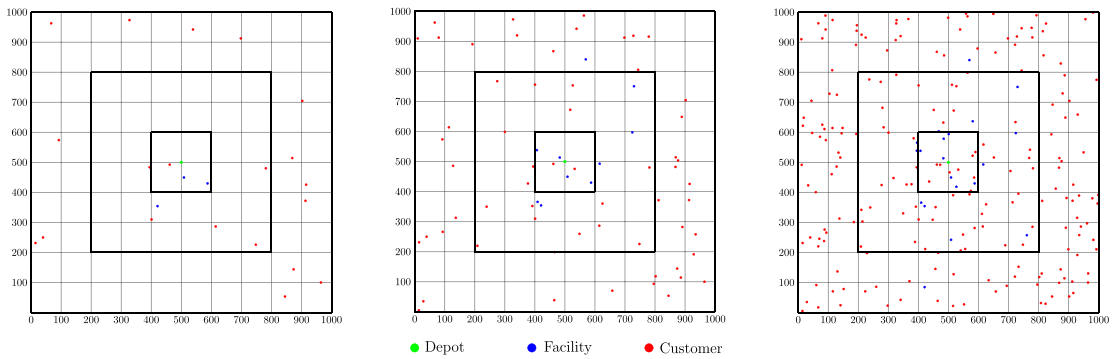
Some conclusions can be observed from the results in Tables 1 and 2.

- Only with the Third Formulation is possible to get optimal solutions ($GAP < 0.1\%$) in the case the number of facilities and drones are small (case $|F| = 3$ and $|V| = 4$).
- If the number of customers and facilities is fixed, GAPs increase with the number of drones almost in all cases.
- Even the instances with the smallest number of customers ($|C| = 20$) generate high GAPs ($GAP > 10\%$) when $|F| = 10$ and $|V| = 8$ (Table 1).
- For the instances with $|C| = 20$ to $|C| = 100$, the Third Formulation gets solutions with a small GAP (< 2.9) for $|F| = 3$.
- The results of the First and Second Formulations are similar. Given the structure of their mathematical models, it is something expected.
- It is important to note that the Third Formulation explicitly uses the number of facilities, while the First and Second Formulations models do not. Therefore, the number of variables and constraints is greater in the Third Formulation than in the other models. This fact justifies the increase of GAPs according to the number of facilities when the number of customers increases from 60 to 100, and 100 to 150.
- The results in Tables 1 and 2 show how GAPs increase as the number of facilities and drones are more significant, and the number of customers is fixed for each model. However, when the number of facilities and drones is fixed, the GAP obtained when the number of customers increases shows only light changes. When $|C| = 150$, the case is different since the First and Second Formulations could not get feasible solutions for several instances. The Third Formulation got feasible solutions for all the instances with 150 customers but achieved very high GAPs.
- For several instances, even with 20 customers, GAPs with values over 10% are achieved. For the instances with 60 and 100 customers and using 8 drones, most results show GAPs with values greater than 10%. Being PDDRP an NP-Hard problem, Tables 1 and 2 clearly show the difficulty of getting optimal solutions for the three formulations. and even solutions with small GAPs.
- An important issue is that lower bounds for the optimal value were obtained for all the test instances, including those with 150 customers. These bounds will help to measure the quality of the GRASP results.

Next, it is shown in detail the results obtained by the GRASP approach.

5.2.2. Results of the GRASP approach

Fig. 6 shows the four GRASP variants obtained by combining the two constructive procedures with the two local search procedures already defined. So, G_1 is the variant including First Constructive and First Local Search, G_2 includes First Constructive and Second Local Search, G_3 includes Second Constructive and First Local Search, and G_4 includes Second Constructive and Second Local Search. For G_1 and



(a) $20c - 3f - I_1$ (b) $60c - 10f - I_1$ (c) $150c - 20f - I_1$

Fig. 5. Three network configurations.

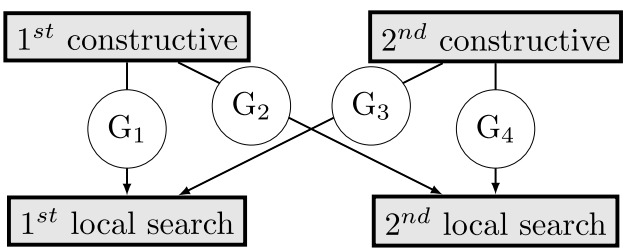


Fig. 6. The four GRASP variants G_1, G_2, G_3, and G_4.

G_2, $\alpha \in \{0, 0.2, 0.4, 0.6, 0.8, 1\}$ is used, while for G_3 and G_4 $\%p \in \{0, 0.2, 0.4, 0.6, 0.8, 1\}$ is used. Each GRASP variant is implemented with $MaxIterations = 100$, for each value of the parameters α and $\%p$. Thus, each one instance of the problem is solved 600 times in every GRASP variant. Once an instance is already solved, the reported result corresponds to the one that induces the best value of w. Experiments show that the best values of α are 0.2, 0, and 0.2 for the First Constructive procedure, G_1 and G_2, respectively. Similarly, the best values of $\%p$ are 0.2, 0.4, and 0.4 for the Second Constructive procedure, G_3 and G_4, respectively. Note that the best solution found by the constructive procedures is not necessarily related to the best solution found after applying the local search procedures.

Table 3 shows a comparison between First and Second Constructive schemes for both groups of instances I_1 and I_2. Then, Tables 4 and 5 show the results for each GRASP scheme, using all combinations of both constructive and local search phases for I_1 and I_2, respectively. Let $G(I)$ be the solution's value founded by GRASP and $UB'(I)$ the value of the best solution found by the three exact models for the instance I. Here the GAPs are calculated as follows:

$$GAP_H(I) = \frac{G(I) - UB'(I)}{UB'(I)} \tag{36}$$

Several findings can be noted from the results in Tables 3–5. For the constructive procedures, Table 3 shows that the First Constructive and the Second Constructive procedure achieves a maximum GAP of 13.94% and 23.21%, respectively, when are compared with the best solution of a MILP formulation. More interesting is that the two methods, for 100 and 150 customers, in most instances, found better solutions than the best solution provided by the exact algorithms.

Concerning the performance of GRASP, there are two challenges. One of them is to know the local search procedures' capacity to improve the results provided by the First and Second Constructive procedures. The other is to understand how GRASP, with its two phases, can improve the best result of the exact algorithms. The conclusions from Tables 4 and 5 for these challenges are commented below.

- All GRASP variants obtained a good performance, noting that running times are not very different for a fixed number of customers.
- In the four variants of GRASP, the local search phase was able to improve the results provided by the constructive phase strongly. In particular, for 40 customers, all GRASP versions outperformed exact algorithms in some parameter combinations. Furthermore, for 60, 100, and 150 customers, GRASP versions achieved better solutions than exact approaches for most instances with 10 or more facilities.
- Variant G_4 achieved the best performance among the GRASP variants. It obtained 20% of improvement for the best exact approach in two instances of I_1, as shown in Table 4. It also occurs in four instances of I_2, as is presented in Table 5. Additionally, the running times of this version are small. As it is shown in Tables 4 and 5, the algorithm takes only 565.5 s as its maximum running time.

5.3. Modeling and algorithmic findings

Some findings have emerged after the computational experiments with the set of problem instances. The first interesting issue is the performance of the exact algorithms. The First and Second Formulations find feasible solutions with a GAP lower than 30% in the two groups of instances with up to 100 customers. For most instances in both groups, the Third Formulation finds the best solution. However, for the case of a fleet of 8 drones, this approach obtained its worst result, a GAP of 71.59%, for instance, in Group I_1, with 100 customers, 10 facilities, and 8 drones. The complexity of the problem can also be appreciated in the instances with 150 customers in Tables 1 and 2. Both the First and Second Formulations could not get a feasible solution within one hour's time limit. The Third Formulation brings feasible solutions for all the instances, but GAPs more significant than 50% were achieved in a few cases. Thus high GAPs are achieved by exact approaches for several instances, principally for 100 and 150 customers. A positive experience with the results of the exact algorithms is lower bounds of the optimum were obtained for all the instances. These bounds will help evaluate the quality of feasible solutions obtained by the GRASP variants.

Some interesting findings are found when the results from both exact and GRASP algorithms were analyzed in an integrated form. Table 6 shows the Best Known Solutions (BKS) for the two groups of instances. The table also shows the value of the best lower bound of a feasible solution found by an exact algorithm, the GAP between the two previous values, and the algorithm(s) achieving these best values. This GAP is obtained using Eq. (35). In this case, UB(I) is the objective function value obtained with the different solution methods, and LB(I) is the best lower bound obtained with the exact procedures. Two facts are important: the first is for small instances with until 60 customers; the Third Formulation achieved the most of the best

Table 1
Results of exact approaches for Group 1 of Instances.

Instance	1st Formulation			2nd Formulation			3rd Formulation		
	w ($units$)	t (s)	GAP (%)	w ($units$)	t (s)	GAP (%)	w ($units$)	t (s)	GAP (%)
$20c-3f-4v$	3626.19	3600	2.89	3629.45	3600	3.46	3602.06	295	0.01
$20c-3f-6v$	2469.20	3600	3.40	2479.11	3600	5.73	2469.20	3600	0.89
$20c-3f-8v$	1942.56	3600	5.46	1969.97	3600	11.06	1933.46	3600	2.86
$20c-5f-4v$	3154.77	3600	2.20	3166.62	3600	7.18	3154.77	2888	0.01
$20c-5f-6v$	2215.96	3600	4.40	2215.96	3600	11.54	2205.20	3600	2.12
$20c-5f-8v$	1774.85	3600	8.08	1786.83	3600	16.38	1774.85	3600	5.08
$20c-10f-4v$	2905.96	3600	3.34	2905.96	3600	10.04	2904.57	3600	3.04
$20c-10f-6v$	2022.67	3600	2.32	2022.67	3600	12.62	2096.62	3600	9.20
$20c-10f-8v$	1707.74	3600	10.92	1702.29	3600	17.52	1896.36	3600	22.43
$20c-15f-4v$	2494.78	3600	8.15	2474.44	3600	16.23	2550.79	3600	13.58
$20c-15f-6v$	1850.28	3600	13.79	1820.25	3600	21.64	1982.60	3600	22.04
$20c-15f-8v$	1598.47	3600	20.65	1598.47	3600	20.65	1667.28	3600	25.88
$40c-3f-4v$	7196.88	3600	1.22	7241.61	3600	1.87	7141.50	3600	0.05
$40c-3f-6v$	4885.26	3600	2.94	4904.25	3600	3.40	4789.50	3600	0.25
$40c-3f-8v$	3716.21	3600	4.22	3796.30	3600	6.40	3636.22	3600	0.93
$40c-5f-4v$	5984.89	3600	2.74	6017.33	3600	4.81	5903.75	3600	0.11
$40c-5f-6v$	4143.24	3600	4.91	4292.97	3600	11.05	4022.93	3600	0.48
$40c-5f-8v$	3198.40	3600	6.16	3339.30	3600	14.23	3135.53	3600	3.57
$40c-10f-4v$	4493.83	3600	4.58	4677.93	3600	10.11	4479.40	3600	3.48
$40c-10f-6v$	3277.37	3600	10.88	3448.13	3600	18.70	3246.78	3600	9.76
$40c-10f-8v$	2526.99	3600	11.35	2717.20	3600	22.63	2764.11	3600	18.62
$40c-15f-4v$	4476.82	3600	7.21	4544.09	3600	10.51	4474.24	3600	6.71
$40c-15f-6v$	3352.52	3600	15.32	3299.68	3600	17.98	3181.20	3600	10.40
$40c-15f-8v$	2571.70	3600	15.15	2717.17	3600	25.20	2804.91	3600	21.79
$60c-3f-4v$	10619.74	3600	1.77	10616.52	3600	1.74	10456.90	3600	0.02
$60c-3f-6v$	7218.63	3600	3.65	7184.66	3600	3.21	6988.51	3600	0.10
$60c-3f-8v$	5506.19	3600	5.25	5467.94	3600	4.61	5265.49	3600	0.27
$60c-5f-4v$	8792.23	3600	3.09	8915.92	3600	5.27	8578.80	3600	0.05
$60c-5f-6v$	6164.39	3600	7.02	6081.38	3600	7.41	5804.46	3600	1.16
$60c-5f-8v$	4649.88	3600	6.69	4760.06	3600	11.36	4408.93	3600	1.49
$60c-10f-4v$	8262.41	3600	7.11	8199.98	3600	6.94	7834.18	3600	1.69
$60c-10f-6v$	5577.45	3600	7.28	5636.65	3600	9.74	5586.77	3600	7.29
$60c-10f-8v$	4372.94	3600	10.30	4575.91	3600	16.62	4437.93	3600	11.47
$60c-15f-4v$	6713.78	3600	6.92	6755.79	3600	8.63	6650.99	3600	5.87
$60c-15f-6v$	4879.39	3600	13.47	4821.79	3600	14.69	4725.17	3600	10.44
$60c-15f-8v$	3854.01	3600	16.60	3952.90	3600	22.03	3598.52	3600	10.31
$100c-3f-4v$	17222.59	3600	1.39	17128.30	3600	0.85	17003.79	3600	0.01
$100c-3f-6v$	11620.58	3600	2.56	11633.97	3600	2.68	11347.36	3600	0.04
$100c-3f-8v$	8869.34	3600	4.25	9309.19	3600	8.78	8527.38	3600	0.13
$100c-5f-4v$	14410.72	3600	3.65	14275.85	3600	3.01	13949.17	3158	0.01
$100c-5f-6v$	9866.78	3600	5.90	9988.38	3600	7.59	9361.41	3600	0.63
$100c-5f-8v$	7581.70	3600	7.70	7750.64	3600	10.68	7110.97	3600	1.55
$100c-10f-4v$	13445.45	3600	7.19	12989.21	3600	4.12	12589.35	3600	0.64
$100c-10f-6v$	9371.25	3600	10.93	9318.39	3600	11.00	9158.42	3600	8.68
$100c-10f-8v$	7097.29	3600	11.31	7184.02	3600	13.42	22181.80	3600	71.59
$100c-15f-4v$	11374.14	3600	9.69	11177.10	3600	8.48	10560.54	3600	2.54
$100c-15f-6v$	7873.41	3600	12.67	8062.32	3600	15.38	11588.48	3600	40.55
$100c-15f-8v$	6330.47	3600	17.90	6247.30	3600	18.41	7308.21	3600	28.80
$150c-5f-4v$	21214.41	3600	1.85	–	3600	–	20864.86	2472	0.01
$150c-5f-6v$	14912.02	3600	6.76	–	3600	–	13972.99	3600	0.43
$150c-5f-8v$	–	3600	–	–	3600	–	11739.56	3600	10.94
$150c-5f-12v$	–	3600	–	–	3600	–	7380.00	3600	4.94
$150c-10f-4v$	20620.77	3600	7.32	20125.43	3600	5.20	19384.40	3600	1.34
$150c-10f-6v$	13872.40	3600	8.03	13608.24	3600	6.53	48743.41	3600	73.80
$150c-10f-8v$	11567.54	3600	17.12	11583.67	3600	17.65	22847.02	3600	58.01
$150c-10f-12v$	–	3600	–	7643.64	3600	16.80	16625.65	3600	61.27
$150c-15f-4v$	17183.87	3600	7.92	17337.48	3600	9.05	17789.89	3600	10.92
$150c-15f-6v$	12811.66	3600	17.46	12373.7	3600	15.05	39658.37	3600	73.30
$150c-15f-8v$	–	3600	–	10287.97	3600	23.37	2010915.14	3600	99.60
$150c-15f-12v$	8350.51	3600	45.58	–	3600	–	28234.87	3600	80.97
$150c-20f-4v$	16042.10	3600	9.89	15961.05	3600	9.80	119994.00	3600	87.93
$150c-20f-6v$	11431.16	3600	15.47	11337.92	3600	15.35	31541.86	3600	69.31
$150c-20f-8v$	–	3600	–	8389.11	3600	14.19	117925.90	3600	93.81
$150c-20f-12v$	–	3600	–	6105.15	3600	21.39	114548.00	3600	95.70

solutions. Besides, for the larger instances with 60 customers, version G_4 of GRASP got the best solution for the two groups of instances. In the case of the larger groups of instances with 100 and 150 customers, the G_4 GRASP variant got the most of the best solutions. It achieved relatively small GAPs when considering the lower bounds provided by the exact algorithms. Furthermore, Third Formulation obtain the best solution for the smaller instance with 100 and 150 customers.

To get a good look at the quality solutions, in Figs. 7–11, a comparison between different GRASP versions and mathematical models is carried out. In these figures, axis X represents the instances of I_1 analyzed, and axis Y represents the GAP with respect to the lower obtained with CPLEX in the different solutions. This lower bound is the lowest value among all mathematical models executed. Fig. 7 shows a comparison among all mathematical models, where it is possible to

Table 2
Results of exact algorithms for Group 2 of Instances.

Instance	1st Formulation			2nd Formulation			3rd Formulation		
	w	t	GAP	w	t	GAP	w	t	GAP
	(*units*)	(s)	(%)	(*units*)	(s)	(%)	(*units*)	(s)	(%)
$20c-3f-4v$	3203.61	3600	3.66	3204.35	3600	4.00	3170.77	23	0.01
$20c-3f-6v$	2225.99	3600	7.04	2199.01	3600	6.69	2199.01	3600	1.18
$20c-3f-8v$	1738.43	3600	10.22	1738.43	3600	11.52	1738.43	3600	2.70
$20c-5f-4v$	2860.48	3600	3.29	2860.48	3600	5.45	2852.86	209	0.01
$20c-5f-6v$	2044.96	3600	6.47	2044.96	3600	12.01	2044.96	3600	3.48
$20c-5f-8v$	1642.03	3600	8.01	1665.82	3600	14.93	1665.82	3600	10.53
$20c-10f-4v$	2464.43	3600	4.21	2464.43	3600	10.72	2453.35	3600	3.26
$20c-10f-6v$	1799.08	3600	6.71	1800.08	3600	16.18	1808.20	3600	10.84
$20c-10f-8v$	1559.56	3600	13.79	1559.56	3600	26.55	1637.05	3600	22.68
$20c-15f-4v$	2284.98	3600	8.66	2284.98	3600	14.81	2284.98	3600	9.55
$20c-15f-6v$	1668.28	3600	8.11	1668.28	3600	19.49	1747.49	3600	17.31
$20c-15f-8v$	1471.49	3600	18.09	1471.49	3600	27.85	1498.51	3600	23.53
$40c-3f-4v$	6083.88	3600	1.95	6178.01	3600	3.45	6031.85	3600	0.05
$40c-3f-6v$	4154.77	3600	4.28	4295.21	3600	7.42	4067.19	3600	0.30
$40c-3f-8v$	3179.13	3600	6.13	3197.51	3600	6.73	3112.83	3600	1.18
$40c-5f-4v$	5279.09	3600	2.69	5424.02	3600	6.02	5244.44	1887	0.01
$40c-5f-6v$	3736.82	3600	7.18	3757.58	3600	9.56	3561.43	3600	0.47
$40c-5f-8v$	2872.32	3600	8.07	2929.69	3600	13.00	2838.74	3600	6.26
$40c-10f-4v$	4276.89	3600	5.65	4458.09	3600	12.53	4206.74	3600	2.83
$40c-10f-6v$	3119.85	3600	11.39	3275.90	3600	20.65	2966.99	3600	6.19
$40c-10f-8v$	2407.84	3600	11.41	2537.61	3600	23.17	2629.46	3600	18.48
$40c-15f-4v$	3927.38	3600	7.81	4011.42	3600	12.24	3929.53	3600	6.31
$40c-15f-6v$	2778.16	3600	10.45	2994.82	3600	21.63	2966.99	3600	14.99
$40c-15f-8v$	2270.25	3600	15.18	2551.16	3600	31.00	2383.49	3600	18.21
$60c-3f-4v$	9338.88	3600	1.15	9516.18	3600	2.99	9279.35	185	0.01
$60c-3f-6v$	6335.07	3600	2.85	6445.03	3600	4.51	6216.73	3600	0.18
$60c-3f-8v$	4920.23	3600	6.18	4987.76	3600	7.46	4695.51	3600	0.59
$60c-5f-4v$	8042.91	3600	3.67	8060.33	3600	4.13	7839.75	3600	0.13
$60c-5f-6v$	5634.66	3600	7.82	5647.17	3600	8.78	5294.96	3600	0.43
$60c-5f-8v$	4328.38	3600	9.37	4433.99	3600	12.86	4077.99	3600	3.29
$60c-10f-4v$	6655.01	3600	9.28	6622.32	3600	9.74	6267.04	3600	2.73
$60c-10f-6v$	4795.59	3600	14.85	4791.74	3600	16.83	4464.38	3600	7.93
$60c-10f-8v$	3579.67	3600	13.03	4006.52	3600	25.44	3512.69	3600	10.91
$60c-15f-4v$	6514.35	3600	17.26	6127.54	3600	12.95	5801.51	3600	6.13
$60c-15f-6v$	4718.76	3600	22.68	4525.74	3600	21.73	4014.96	3600	8.19
$60c-15f-8v$	3694.70	3600	24.56	3731.93	3600	28.81	3953.17	3600	29.00
$100c-3f-4v$	16003.62	3600	1.37	15928.83	3600	0.91	15803.28	75	0.01
$100c-3f-6v$	10795.33	3600	2.52	10761.24	3600	2.21	10540.92	3600	0.02
$100c-3f-8v$	8408.65	3600	6.14	8248.73	3600	4.32	7923.13	3600	0.20
$100c-5f-4v$	14002.11	3600	4.25	13788.12	3600	2.79	13461.75	3600	0.04
$100c-5f-6v$	9419.39	3600	5.05	9454.92	3600	5.49	9028.80	3600	0.77
$100c-5f-8v$	7433.41	3600	9.64	7469.55	3600	10.28	6820.16	3600	1.29
$100c-10f-4v$	11560.51	3600	10.00	11152.69	3600	7.05	10783.76	3600	3.14
$100c-10f-6v$	7983.09	3600	12.70	7922.83	3600	12.81	7265.52	3600	3.77
$100c-10f-8v$	6306.77	3600	16.68	6637.71	3600	21.95	5638.13	3600	6.52
$100c-15f-4v$	10617.14	3600	14.27	10117.24	3600	10.50	9640.96	3600	5.11
$100c-15f-6v$	7364.10	3600	17.05	7359.04	3600	18.13	6906.02	3600	11.21
$100c-15f-8v$	5854.83	3600	21.15	5762.60	3600	21.59	5688.37	3600	18.58
$150c-5f-4v$	20161.22	3600	2.43	20339.57	3600	3.28	19709.47	3398	0.01
$150c-5f-6v$	14116.72	3600	7.10	–	3600	–	13201.04	3600	0.58
$150c-5f-8v$	–	3600	–	–	3600	–	10800.22	3600	8.83
$150c-5f-12v$	–	3600	–	–	3600	–	8108.59	3600	18.88
$150c-10f-4v$	16320.78	3600	8.15	16290.78	3600	7.93	15268.20	3600	1.53
$150c-10f-6v$	11395.06	3600	12.30	11489.89	3600	12.85	24632.70	3600	59.27
$150c-10f-8v$	9235.67	3600	18.84	–	3600	–	29444.93	3600	74.39
$150c-10f-12v$	–	3600	–	–	3600	–	7879.92	3600	35.76
$150c-15f-4v$	14841.97	3600	9.82	14844.56	3600	9.60	90883.97	3600	85.21
$150c-15f-6v$	11117.56	3600	19.74	11220.69	3600	20.07	39066.56	3600	77.01
$150c-15f-8v$	8814.36	3600	24.07	10846.13	3600	37.76	1977672.14	3600	99.66
$150c-15f-12v$	–	3600	–	–	3600	–	29440.04	3600	84.55
$150c-20f-4v$	14269.27	3600	13.53	14512.39	3600	14.69	35353.22	3600	64.92
$150c-20f-6v$	12112.89	3600	32.12	10137.72	3600	18.30	114131.00	3600	92.73
$150c-20f-8v$	7642.99	3600	19.32	–	3600	–	20826.31	3600	69.98
$150c-20f-12v$	–	3600	–	–	3600	–	28957.51	3600	85.44

observe that the Third Formulation achieves better solutions in most cases. Moreover, for the largest instances with 100 and 150 customers, the Third Formulation is the only one to provide feasible solutions but gets high GAPs. In Fig. 8 a comparison between first and the second constructive procedure is carried out. In this figure is possible to observe that the First Constructive procedure normally outperformed the Second for instances with 20 and 40 customers. However, for instances with 60 or more customers, the Second Constructive procedure outperformed the First one. Figs. 9 and 10 show the impact of the local search procedure on the quality of the solutions when both the

Table 3
Results of Constructive phase for Groups 1 and 2 of instances.

Instance	Group 1 of Instances						Group 2 of Instances					
	First Constructive			Second Constructive			First Constructive			Second Constructive		
	w (*units*)	**t** (s)	GAP_H (%)	w (*units*)	**t** (s)	GAP_H (%)	w (*units*)	**t** (s)	GAP_H (%)	w (unit)	**t** (s)	GAP_H (%)
$20c-3f-4v$	3816.55	0.3	5.95	4030.50	0.2	11.89	3308.24	0.2	4.34	3380.08	0.2	6.60
$20c-3f-6v$	2709.57	0.3	9.73	3037.61	0.2	23.02	2446.52	0.4	11.26	2500.23	0.2	13.70
$20c-3f-8v$	2097.48	0.6	8.48	2381.70	0.4	23.18	1950.14	0.7	12.18	1947.96	0.3	12.05
$20c-5f-4v$	3467.22	0.2	9.90	3495.83	0.1	10.81	3132.80	0.2	9.81	3074.96	0.2	7.79
$20c-5f-6v$	2493.71	0.3	13.08	2595.85	0.2	17.71	2215.00	0.3	8.32	2291.60	0.2	12.06
$20c-5f-8v$	2022.26	0.5	13.94	2186.88	0.4	23.21	1847.64	0.6	12.52	1923.82	0.4	17.16
$20c-10f-4v$	3106.25	0.2	6.94	3178.23	0.2	9.42	2681.70	0.2	9.31	2750.05	0.1	12.09
$20c-10f-6v$	2303.19	0.3	13.87	2276.91	0.3	12.57	2015.64	0.4	12.04	2087.69	0.2	16.04
$20c-10f-8v$	1917.75	0.5	12.66	1993.45	0.4	17.10	1758.97	0.6	12.79	1751.31	0.4	12.30
$20c-15f-4v$	2600.13	0.2	5.08	2596.27	0.1	4.92	2390.67	0.2	4.63	2393.82	0.2	4.76
$20c-15f-6v$	2042.70	0.3	12.22	2061.70	0.2	13.26	1875.68	0.4	12.43	1867.05	0.2	11.91
$20c-15f-8v$	1724.16	0.5	7.86	1834.57	0.4	14.77	1617.17	0.6	9.90	1758.58	0.4	19.51
$40c-3f-4v$	7505.97	1.1	5.10	7516.28	0.9	5.25	6399.40	1.2	6.09	6377.39	1.0	5.73
$40c-3f-6v$	5059.98	2.1	5.65	5269.49	1.2	10.02	4458.34	2.2	9.62	4509.38	1.2	10.87
$40c-3f-8v$	4031.50	3.5	10.87	4225.13	1.9	16.20	3397.98	3.8	9.16	3471.53	2.0	11.52
$40c-5f-4v$	6395.79	1.0	8.33	6240.41	0.7	5.70	5705.26	1.1	8.79	5617.02	0.7	7.10
$40c-5f-6v$	4502.54	2.0	11.92	4345.63	1.2	8.02	3940.67	2.2	10.65	4019.10	1.2	12.85
$40c-5f-8v$	3417.71	3.6	9.00	3439.09	1.9	9.68	3090.62	3.8	8.87	3170.06	2.0	11.67
$40c-10f-4v$	4966.99	1.0	10.89	4833.71	0.7	7.91	4681.18	1.1	11.28	4605.21	0.7	9.47
$40c-10f-6v$	3576.83	2.1	10.17	3572.46	1.2	10.03	3300.16	2.2	11.23	3394.11	1.2	14.40
$40c-10f-8v$	2860.37	3.6	13.19	2878.29	2.0	13.90	2724.41	3.9	13.15	2765.09	2.1	14.84
$40c-15f-4v$	4945.39	1.0	10.53	4739.92	0.7	5.94	4268.47	1.0	8.68	4245.73	0.7	8.11
$40c-15f-6v$	3506.83	2.0	10.24	3418.70	1.2	7.47	3111.23	2.2	11.99	3111.68	1.3	12.01
$40c-15f-8v$	2778.84	3.6	8.05	2759.75	2.0	7.31	2547.43	3.9	12.21	2403.72	2.1	5.88
$60c-3f-4v$	10909.68	3.2	4.33	10902.00	2.2	4.26	9794.57	3.5	5.55	9740.10	2.2	4.97
$60c-3f-6v$	7492.93	6.7	7.22	7461.69	3.7	6.77	6607.52	7.1	6.29	6771.40	3.8	8.92
$60c-3f-8v$	5722.20	11.7	8.67	5779.19	6.0	9.76	5245.46	12.5	11.71	5059.74	6.2	7.76
$60c-5f-4v$	9173.24	3.2	6.93	8928.75	1.8	4.08	8419.49	3.3	7.39	8370.72	1.9	6.77
$60c-5f-6v$	6317.88	6.8	8.85	6289.68	3.7	8.36	5833.53	7.3	10.17	5771.46	3.8	9.00
$60c-5f-8v$	4969.37	12.1	12.71	4894.26	6.1	11.01	4502.88	12.7	10.42	4459.98	6.3	9.37
$60c-10f-4v$	8567.83	3.2	9.36	8167.49	1.9	4.25	6817.17	3.3	8.78	6802.38	1.9	8.54
$60c-10f-6v$	5782.85	6.7	3.68	5723.95	3.7	2.63	4794.80	7.3	7.40	4875.25	3.7	9.20
$60c-10f-8v$	4572.10	12.1	4.55	4402.50	6.1	0.68	3878.79	12.9	10.42	3812.48	6.4	8.53
$60c-15f-4v$	7023.60	3.0	5.60	6941.60	1.9	4.37	6149.55	3.2	6.00	6110.54	1.8	5.33
$60c-15f-6v$	5003.15	6.9	5.88	4892.48	3.6	3.54	4530.81	7.1	12.85	4409.92	3.8	9.84
$60c-15f-8v$	3896.73	12.0	8.29	3854.70	6.0	7.12	3527.80	12.9	−4.52	3605.37	6.3	−2.42
$100c-3f-4v$	17530.79	14.7	3.10	17394.87	7.9	2.30	16420.17	15.2	3.90	16230.85	7.8	2.71
$100c-3f-6v$	12076.12	31.7	6.42	11834.31	14.7	4.29	11197.83	32.7	6.23	11032.64	15.1	4.66
$100c-3f-8v$	9156.92	55.8	7.38	9025.06	25.0	5.84	8407.16	57.2	6.11	8464.49	25.3	6.83
$100c-5f-4v$	14736.24	14.6	5.64	14338.70	7.4	2.79	14343.10	14.8	6.55	13947.17	7.4	3.61
$100c-5f-6v$	10007.06	31.3	6.90	9810.40	14.8	4.80	9795.09	32.2	8.49	9556.64	14.8	5.85
$100c-5f-8v$	7770.21	56.2	9.27	7604.90	25.9	6.95	7498.69	57.4	9.95	7379.97	25.3	8.21
$100c-10f-4v$	13538.36	14.1	7.54	13087.37	7.2	3.96	11435.32	14.6	6.04	11102.03	7.4	2.95
$100c-10f-6v$	9281.64	31.3	1.35	8970.77	14.7	−2.05	7680.63	31.9	5.71	7637.99	14.6	5.13
$100c-10f-8v$	7087.19	55.7	−0.14	6879.64	26.6	−3.07	6052.47	57.4	7.35	5942.26	25.5	5.39
$100c-15f-4v$	11398.75	14.0	7.94	10981.48	7.3	3.99	10029.80	14.5	4.03	9746.87	7.1	1.10
$100c-15f-6v$	7752.97	31.1	−1.53	7445.30	14.5	−5.44	7114.27	32.0	3.02	6828.18	14.5	−1.13
$100c-15f-8v$	6028.68	57.2	−3.50	5914.04	26.0	−5.33	5532.26	57.6	−2.74	5359.50	25.9	−5.78
$150c-5f-4v$	21683.24	45.7	3.92	21340.15	21.4	2.28	20388.60	46.2	3.45	20234.67	25.7	2.66
$150c-5f-6v$	15017.51	98.3	7.48	14458.53	44.6	3.47	13815.67	98.1	4.66	13669.31	55.5	3.55
$150c-5f-8v$	11306.26	173.1	−3.69	11063.03	73.0	−5.76	10606.28	171.1	−1.80	10467.27	102.8	−3.08
$150c-5f-12v$	7863.19	417.7	6.55	7602.23	218.8	3.01	7327.18	396.6	−9.64	7253.20	154.4	−10.55
$150c-10f-4v$	20161.29	46.6	4.01	19789.29	21.2	2.09	16088.51	46.4	5.37	15645.92	26.7	2.47
$150c-10f-6v$	13871.87	98.3	1.94	13429.72	42.6	−1.31	11128.44	98.6	−2.34	10736.05	56.1	−5.78
$150c-10f-8v$	10545.77	174.9	−8.83	10252.65	74.1	−11.37	8307.74	170.8	−10.05	8191.93	103.6	−11.30
$150c-10f-12v$	7331.89	411.3	−4.08	7067.74	208.3	−7.53	5907.24	391.3	−25.03	5809.14	155.2	−26.28
$150c-15f-4v$	16894.64	47.0	−1.68	16489.44	21.1	−4.04	14222.75	46.7	−4.17	14068.71	24.9	−5.21
$150c-15f-6v$	11622.77	101.1	−6.07	11245.26	42.7	−9.12	9843.43	98.1	−11.46	9710.77	55.6	−12.65
$150c-15f-8v$	8814.95	175.1	−14.32	8635.06	74.0	−16.07	7622.95	172.7	−13.52	7516.50	98.6	−14.72
$150c-15f-12v$	6171.15	429.6	−26.10	6094.10	218.3	−27.02	5392.43	386.7	−81.68	5346.30	152.3	−81.84
$150c-20f-4v$	15541.69	46.9	−2.63	15310.04	21.1	−4.08	13265.45	46.0	−7.03	13148.67	24.6	−7.85
$150c-20f-6v$	10799.23	99.6	−4.75	10456.25	42.7	−7.78	9202.10	97.4	−9.23	9071.09	57.5	−10.52
$150c-20f-8v$	8348.61	177.2	−0.48	8043.40	73.6	−4.12	7161.90	171.7	−6.29	7019.09	99.2	−8.16
$150c-20f-12v$	5921.02	425.8	−3.02	5723.16	212.2	−6.26	5185.42	390.0	−82.09	5030.66	148.9	−82.63

First and Second Constructive procedures are executed in the first phase of GRASP, respectively. Both local searches achieved a good performance, but the Second Constructive is superior. Finally, in Fig. 11 a comparison between the best solution obtained in each instance with the mathematical models and the GRASP versions is carried out. In this figure is possible to observe that until most instances with 60 customers, the best solution given by an exact formulation provides solutions with a GAP relatively low. At the same time, the best solution given by a GRASP variant is very near to the exact solution and, sometimes, it improves this value (e.g., largest instances with 40 customers). However, from the largest instances with 60 customers, the G_4 version outperformed exact algorithms in most cases. Moreover,

Table 4
GRASP results for Group 1 of Instances.

Instance	G_1			G_2			G_3			G_4		
	w (*units*)	**t** (s)	GAP_H (%)	**w** (*units*)	**t** (s)	GAP_H (%)	**w** (*units*)	**t** (s)	GAP_H (%)	**w** (*units*)	**t** (s)	GAP_H (%)
$20c-3f-4v$	3657.60	0.4	1.54	3642.58	0.5	1.12	3649.89	0.3	1.33	3649.89	0.5	1.33
$20c-3f-6v$	2533.03	0.4	2.59	2532.62	0.4	2.57	2536.96	0.3	2.74	2536.89	0.3	2.74
$20c-3f-8v$	1983.47	0.6	2.59	1983.47	0.6	2.59	1947.24	0.4	0.71	1947.24	0.4	0.71
$20c-5f-4v$	3240.83	0.2	2.73	3172.31	0.3	0.56	3212.48	0.2	1.83	3207.92	0.3	1.68
$20c-5f-6v$	2282.07	0.4	3.49	2226.30	0.4	0.96	2285.30	0.3	3.63	2294.71	0.3	4.06
$20c-5f-8v$	1882.78	0.6	6.08	1858.82	0.6	4.73	1882.78	0.4	6.08	1869.70	0.4	5.34
$20c-10f-4v$	2905.96	0.2	0.05	2905.96	0.3	0.05	2956.49	0.2	1.79	2930.59	0.3	0.90
$20c-10f-6v$	2114.81	0.4	4.56	2107.28	0.4	4.18	2072.92	0.3	2.48	2094.75	0.3	3.56
$20c-10f-8v$	1706.77	0.6	0.26	1705.15	0.6	0.17	1768.18	0.4	3.87	1740.41	0.4	2.24
$20c-15f-4v$	2533.41	0.2	2.38	2522.16	0.3	1.93	2537.97	0.2	2.57	2495.97	0.3	0.87
$20c-15f-6v$	1891.33	0.4	3.90	1831.77	0.4	0.63	1878.73	0.3	3.21	1850.28	0.3	1.65
$20c-15f-8v$	1598.47	0.6	0.00	1598.47	0.6	0.00	1635.99	0.4	2.35	1598.47	0.4	0.00
$40c-3f-4v$	7300.03	1.3	2.22	7194.51	2.6	0.74	7215.28	1.0	1.03	7178.75	2.2	0.52
$40c-3f-6v$	4886.74	2.2	2.03	4850.38	2.6	1.27	4828.77	1.3	0.82	4825.03	1.7	0.74
$40c-3f-8v$	3752.26	3.5	3.19	3730.58	3.8	2.60	3717.66	2.0	2.24	3725.16	2.2	2.45
$40c-5f-4v$	6140.11	1.2	4.00	5984.10	2.3	1.36	6017.25	0.8	1.92	5968.19	1.6	1.09
$40c-5f-6v$	4262.14	2.1	5.95	4145.48	2.6	3.05	4153.71	1.3	3.25	4119.96	1.7	2.41
$40c-5f-8v$	3269.36	3.7	4.27	3211.71	4.0	2.43	3227.94	2.0	2.95	3202.27	2.2	2.13
$40c-10f-4v$	4681.79	1.2	4.52	4547.87	2.5	1.53	4618.85	0.8	3.11	4536.06	1.9	1.26
$40c-10f-6v$	3260.25	2.2	0.41	3203.81	2.7	−1.32	3177.12	1.3	−2.15	3182.74	1.7	−1.97
$40c-10f-8v$	2553.96	3.7	1.07	2453.30	4.0	−2.92	2508.26	2.1	−0.74	2498.27	2.3	−1.14
$40c-15f-4v$	4585.29	1.2	2.48	4463.31	2.6	−0.24	4516.43	0.8	0.94	4416.57	1.9	−1.29
$40c-15f-6v$	3162.85	2.2	−0.58	3154.24	2.6	−0.85	3082.28	1.4	−3.11	3049.42	1.7	−4.14
$40c-15f-8v$	2557.64	3.7	−0.55	2529.57	4.0	−1.64	2458.40	2.1	−4.41	2458.40	2.3	−4.41
$60c-3f-4v$	10626.57	3.4	1.62	10495.49	9.0	0.37	10542.38	2.4	0.82	10484.14	7.0	0.26
$60c-3f-6v$	7167.79	7.0	2.57	7047.82	9.1	0.85	7061.33	3.9	1.04	7022.80	5.6	0.49
$60c-3f-8v$	5365.68	12.0	1.90	5345.12	13.0	1.51	5341.95	6.2	1.45	5304.01	7.0	0.73
$60c-5f-4v$	8933.29	3.5	4.13	8680.12	9.4	1.18	8742.88	2.1	1.91	8641.52	6.7	0.73
$60c-5f-6v$	6059.90	7.0	4.40	5904.31	9.1	1.72	5936.69	4.0	2.28	5877.08	5.6	1.25
$60c-5f-8v$	4559.85	12.4	3.42	4539.98	13.5	2.97	4516.64	6.3	2.44	4490.22	7.2	1.84
$60c-10f-4v$	8119.29	3.6	3.64	7893.28	10.5	0.75	7971.53	2.1	1.75	7860.37	6.9	0.33
$60c-10f-6v$	5476.39	7.0	−1.81	5412.00	9.5	−2.97	5416.30	4.0	−2.89	5351.32	5.9	−4.05
$60c-10f-8v$	4296.00	12.4	−1.76	4162.50	13.7	−4.81	4141.58	6.4	−5.29	4112.86	7.6	−5.95
$60c-15f-4v$	6730.11	3.4	1.19	6523.68	9.8	−1.91	6604.43	2.1	−0.70	6459.99	6.7	−2.87
$60c-15f-6v$	4704.41	7.3	−0.44	4519.81	9.8	−4.35	4575.98	3.9	−3.16	4485.36	5.8	−5.08
$60c-15f-8v$	3596.19	12.4	−0.06	3548.39	13.6	−1.39	3540.05	6.3	−1.62	3462.28	7.4	−3.79
$100c-3f-4v$	17091.75	15.5	0.52	17041.32	54.9	0.22	17084.87	8.6	0.48	17025.54	32.2	0.13
$100c-3f-6v$	11469.63	32.5	1.08	11407.89	47.0	0.53	11439.63	15.5	0.81	11379.49	25.6	0.28
$100c-3f-8v$	8650.46	56.6	1.44	8604.78	63.4	0.91	8608.56	25.8	0.95	8561.83	30.7	0.40
$100c-5f-4v$	14286.16	15.5	2.42	14055.47	55.5	0.76	14063.00	8.2	0.82	14012.19	30.8	0.45
$100c-5f-6v$	9631.83	32.3	2.89	9466.15	47.1	1.12	9500.28	15.6	1.48	9433.89	25.3	0.77
$100c-5f-8v$	7449.69	57.1	4.76	7205.69	64.9	1.33	7242.15	26.7	1.84	7171.59	32.4	0.85
$100c-10f-4v$	12986.47	15.4	3.15	12718.35	68.7	1.02	12838.87	8.1	1.98	12679.52	43.2	0.72
$100c-10f-6v$	8905.37	32.6	−2.76	8613.77	51.4	−5.95	8663.68	15.7	−5.40	8567.39	29.3	−6.45
$100c-10f-8v$	6680.21	56.8	−5.88	6568.17	65.8	−7.46	6574.43	27.5	−7.37	6511.00	34.5	−8.26
$100c-15f-4v$	10752.14	15.6	1.81	10574.60	69.2	0.13	10657.49	8.4	0.92	10460.03	41.6	−0.95
$100c-15f-6v$	7285.55	32.7	−7.47	7141.98	51.0	−9.29	7244.48	15.4	−7.99	7121.67	28.0	−9.55
$100c-15f-8v$	5622.96	58.7	−9.99	5530.19	68.2	−11.48	5530.66	27.0	−11.47	5468.18	34.4	−12.47
$150c-5f-4v$	21270.32	48.7	1.94	20943.49	258.9	0.38	21065.04	23.5	0.96	20937.59	129.7	0.35
$150c-5f-6v$	14279.35	100.9	2.19	14075.72	170.4	0.74	14144.59	46.5	1.23	14050.92	89.0	0.56
$150c-5f-8v$	10881.13	175.4	−7.31	10612.06	210.3	−9.60	10660.98	74.8	−9.19	10580.44	98.9	−9.87
$150c-5f-12v$	7394.48	837.7	0.20	7234.79	853.6	−1.97	7226.06	439.4	−2.09	7177.2	451.1	−2.75
$150c-10f-4v$	19621.22	50.4	1.22	19289.69	354.4	−0.49	19530.92	23.6	0.76	19257.72	188.0	−0.65
$150c-10f-6v$	13334.14	101.5	−2.01	13026.73	195.0	−4.27	13097.61	44.7	−3.75	12931.37	105.8	−4.97
$150c-10f-8v$	10047.53	177.8	−13.14	9862.98	222.9	−14.74	9894.63	76.1	−14.46	9796.44	108.5	−15.31
$150c-10f-12v$	6781.23	825.3	−11.28	6690.22	842.9	−12.47	6707.65	418.6	−12.25	6623.55	431.6	−13.35
$150c-15f-4v$	16353.48	52.1	−4.83	16091.50	384.1	−6.36	16246.45	23.6	−5.46	16040.52	206.0	−6.65
$150c-15f-6v$	11197.04	105.6	−9.51	10892.72	207.6	−11.97	10969.57	45.1	−11.35	10808.37	107.6	−12.65
$150c-15f-8v$	8448.81	179.1	−17.88	8284.96	225.1	−19.47	8328.93	76.4	−19.04	8186.08	109.3	−20.43
$150c-15f-12v$	5758.18	862.9	−31.04	5701.12	882.1	−31.73	5676.16	438.9	−32.03	5615.34	453.8	−32.75
$150c-20f-4v$	15132.93	52.3	−5.19	14743.46	375.8	−7.63	15011.50	23.8	−5.95	14711.77	201.3	−7.83
$150c-20f-6v$	10295.72	104.7	−9.19	10043.57	203.5	−11.42	10180.04	45.4	−10.21	9949.94	107.5	−12.24
$150c-20f-8v$	7866.44	181.9	−6.23	7663.39	227.7	−8.65	7720.95	76.1	−7.96	7582.64	106.1	−9.61
$150c-20f-12v$	5375.94	855.6	−11.94	5311.14	873.9	−13.01	5311.34	427.0	−13.00	5209.33	441.9	−14.67

for most instances with 100 and 150 customers, the exact algorithm presents high GAPs.

5.4. Statistical analysis for evaluating the best version of GRASP

A Wilcoxon signed ranks test was carried out to evaluate what version of GRASP obtain the best out. This test is recommended for evaluating two algorithms (Derrac et al., 2011). The results of these tests are presented in Table 7 where R^+ is the sum of ranks for the instances in which the first algorithm outperformed the second, and R^- the sum of ranks for the opposite. For GRASP variants, a comparison was carried out between G_1 versus G_2 and G_3 versus G_4. The test results with a significance level of less than 0.0001 show that G_2 is better than

Table 5
GRASP results for Group 2 of Instances.

Instance	G_1			G_2			G_3			G_4		
	w (*units*)	**t** (s)	GAP_H (%)	**w** (*units*)	**t** (s)	GAP_H (%)	**w** (*units*)	**t** (s)	GAP_H (%)	**w** (*units*)	**t** (s)	GAP_H (%)
$20c-3f-4v$	3246.36	0.3	2.38	3211.05	0.4	1.27	3232.28	0.4	1.94	3208.63	0.4	1.19
$20c-3f-6v$	2284.49	0.4	3.89	2253.64	0.4	2.48	2269.37	0.3	3.20	2269.37	0.3	3.20
$20c-3f-8v$	1765.34	0.7	1.55	1754.01	0.7	0.90	1806.73	0.4	3.93	1806.73	0.4	3.93
$20c-5f-4v$	2986.78	0.2	4.69	2927.62	0.3	2.62	2979.89	0.2	4.45	2935.57	0.3	2.90
$20c-5f-6v$	2134.91	0.4	4.40	2121.59	0.4	3.75	2136.75	0.3	4.49	2134.91	0.3	4.40
$20c-5f-8v$	1689.92	0.6	2.92	1668.28	0.7	1.60	1684.55	0.4	2.59	1684.55	0.4	2.59
$20c-10f-4v$	2524.85	0.2	2.91	2496.07	0.4	1.74	2505.98	0.2	2.15	2505.98	0.3	2.15
$20c-10f-6v$	1818.75	0.4	1.09	1818.75	0.5	1.09	1809.11	0.3	0.56	1809.11	0.3	0.56
$20c-10f-8v$	1559.56	0.6	0.00	1559.56	0.7	0.00	1559.56	0.4	0.00	1559.56	0.4	0.00
$20c-15f-4v$	2291.15	0.3	0.27	2289.93	0.4	0.22	2284.98	0.2	0.00	2284.98	0.3	0.00
$20c-15f-6v$	1712.37	0.4	2.64	1684.03	0.5	0.94	1729.71	0.3	3.68	1729.71	0.3	3.68
$20c-15f-8v$	1467.48	0.7	−0.27	1467.48	0.7	−0.27	1471.49	0.4	0.00	1471.49	0.4	0.00
$40c-3f-4v$	6261.98	1.3	3.82	6118.95	2.6	1.44	6140.65	1.1	1.80	6082.90	2.2	0.85
$40c-3f-6v$	4255.47	2.3	4.63	4161.82	2.7	2.33	4172.53	1.3	2.59	4145.42	1.7	1.92
$40c-3f-8v$	3245.99	3.9	4.28	3189.65	4.1	2.47	3233.25	2.1	3.87	3196.31	2.3	2.68
$40c-5f-4v$	5483.81	1.2	4.56	5311.13	2.5	1.27	5397.00	0.8	2.91	5326.80	1.8	1.57
$40c-5f-6v$	3778.61	2.3	6.10	3680.58	2.8	3.35	3752.70	1.3	5.37	3681.60	1.7	3.37
$40c-5f-8v$	2934.05	3.9	3.36	2881.40	4.2	1.50	2930.04	2.1	3.22	2860.16	2.4	0.75
$40c-10f-4v$	4453.85	1.3	5.87	4309.46	2.8	2.44	4345.78	0.8	3.31	4307.61	1.9	2.40
$40c-10f-6v$	3076.41	2.4	3.69	3052.52	3.0	2.88	3054.68	1.3	2.96	3023.07	1.8	1.89
$40c-10f-8v$	2454.32	4.0	1.93	2397.53	4.3	−0.43	2436.63	2.3	1.20	2413.87	2.5	0.25
$40c-15f-4v$	4032.84	1.2	2.69	3933.20	2.5	0.15	3982.35	0.8	1.40	3908.56	1.6	−0.48
$40c-15f-6v$	2837.35	2.4	2.13	2783.66	2.9	0.20	2794.53	1.4	0.59	2763.54	1.8	−0.53
$40c-15f-8v$	2297.82	4.1	1.21	2241.60	4.4	−1.26	2243.66	2.2	−1.17	2220.32	2.5	−2.20
$60c-3f-4v$	9384.32	3.8	1.13	9359.53	9.6	0.86	9359.84	2.4	0.87	9330.29	5.9	0.55
$60c-3f-6v$	6381.06	7.4	2.64	6315.79	9.6	1.59	6322.46	4.0	1.70	6290.91	5.5	1.19
$60c-3f-8v$	4835.27	12.8	2.98	4806.24	14.0	2.36	4802.06	6.4	2.27	4782.90	7.4	1.86
$60c-5f-4v$	8064.63	3.6	2.87	7902.18	9.8	0.80	8025.97	2.2	2.38	7909.35	5.9	0.89
$60c-5f-6v$	5503.24	7.6	3.93	5437.25	9.9	2.69	5475.23	4.1	3.40	5390.50	5.6	1.80
$60c-5f-8v$	4286.12	13.0	5.10	4160.78	14.2	2.03	4204.42	6.6	3.10	4126.49	7.6	1.19
$60c-10f-4v$	6470.21	3.8	3.24	6392.27	10.6	2.00	6440.65	2.2	2.77	6301.02	6.3	0.54
$60c-10f-6v$	4436.37	7.7	−0.63	4425.95	10.2	−0.86	4434.58	4.0	−0.67	4368.30	5.8	−2.15
$60c-10f-8v$	3520.20	13.3	0.21	3399.29	14.6	−3.23	3465.99	6.7	−1.33	3422.42	7.8	−2.57
$60c-15f-4v$	5920.53	3.7	2.05	5734.90	10.2	−1.15	5763.83	2.1	−0.65	5667.25	6.0	−2.31
$60c-15f-6v$	4087.26	7.6	1.80	3979.67	10.1	−0.88	4018.74	4.1	0.09	3937.34	5.9	−1.93
$60c-15f-8v$	3218.11	13.4	−12.90	3132.45	14.8	−15.22	3125.05	6.6	−15.42	3103.63	7.7	−16.00
$100c-3f-4v$	15937.17	16.0	0.85	15844.05	54.3	0.26	15886.77	8.5	0.53	15817.75	28.7	0.09
$100c-3f-6v$	10698.50	33.5	1.49	10609.55	48.4	0.65	10613.80	15.9	0.69	10588.11	24.9	0.45
$100c-3f-8v$	8104.64	58.0	2.29	8009.04	65.4	1.08	8016.04	26.0	1.17	7985.92	31.3	0.79
$100c-5f-4v$	13789.01	15.8	2.43	13569.95	58.9	0.80	13631.64	8.1	1.26	13548.54	29.9	0.64
$100c-5f-6v$	9304.42	33.3	3.05	9124.99	49.7	1.07	9154.62	15.6	1.39	9094.05	26.3	0.72
$100c-5f-8v$	6995.01	58.3	2.56	6971.59	66.1	2.22	6968.67	26.1	2.18	6917.40	31.4	1.43
$100c-10f-4v$	10894.90	16.0	1.03	10719.35	66.9	−0.60	10774.76	8.4	−0.08	10612.49	35.8	−1.59
$100c-10f-6v$	7347.18	33.5	1.12	7264.27	51.6	−0.02	7283.52	15.6	0.25	7203.57	27.0	−0.85
$100c-10f-8v$	5673.18	58.8	0.62	5626.44	67.9	−0.21	5602.90	26.4	−0.62	5512.97	32.8	−2.22
$100c-15f-4v$	9596.06	16.2	−0.47	9386.30	66.8	−2.64	9470.58	8.1	−1.77	9328.37	33.2	−3.24
$100c-15f-6v$	6535.41	33.9	−5.37	6470.30	51.8	−6.31	6449.99	15.5	−6.60	6389.74	25.7	−7.48
$100c-15f-8v$	5073.01	59.2	−10.82	4970.94	68.3	−12.61	4962.57	27.0	−12.76	4871.71	33.1	−14.36
$150c-5f-4v$	19968.68	49.4	1.32	19785.47	280.8	0.39	19905.75	28.3	1.00	19773.26	157.2	0.32
$150c-5f-6v$	13440.45	100.7	1.81	13293.72	178.2	0.70	13316.72	57.5	0.88	13243.73	105.5	0.32
$150c-5f-8v$	10126.28	173.3	−6.24	10012.82	210.0	−7.29	10054.81	104.9	−6.90	9993.77	129.8	−7.47
$150c-5f-12v$	6916.68	795.6	−14.70	6825.94	811.9	−15.82	6836.57	552.8	−15.69	6770.42	565.5	−16.50
$150c-10f-4v$	15535.23	50.9	1.75	15274.80	319.5	0.04	15376.07	29.8	0.71	15206.97	193.1	−0.40
$150c-10f-6v$	10459.49	103.0	−8.21	10336.54	191.8	−9.29	10323.56	59.1	−9.40	10239.60	110.0	−10.14
$150c-10f-8v$	7924.07	174.6	−14.20	7808.42	216.9	−15.45	7845.25	106.2	−15.05	7762.05	134.9	−15.96
$150c-10f-12v$	5412.77	786.4	−31.31	5412.51	802.9	−31.31	5381.02	549.0	−31.71	5312.67	560.9	−32.58
$150c-15f-4v$	13861.45	51.8	−6.61	13739.47	334.2	−7.43	13772.05	27.8	−7.21	13619.41	168.4	−8.24
$150c-15f-6v$	9403.97	102.9	−15.41	9289.04	190.2	−16.45	9314.43	58.3	−16.22	9209.42	107.8	−17.16
$150c-15f-8v$	7213.00	177.0	−18.17	7049.37	219.4	−20.02	7098.46	101.3	−19.47	6997.78	129.1	−20.61
$150c-15f-12v$	4998.49	777.6	−83.02	4881.52	795.0	−83.42	4896.75	541.5	−83.37	4805.19	555.0	−83.68
$150c-20f-4v$	12798.43	52.1	−10.31	12612.83	354.8	−11.61	12785.16	27.9	−10.40	12575.35	179.0	−11.87
$150c-20f-6v$	8747.10	103.1	−13.72	8617.74	193.3	−14.99	8631.98	60.6	−14.85	8546.74	112.3	−15.69
$150c-20f-8v$	6718.29	176.8	−12.10	6583.82	221.0	−13.86	6603.96	102.0	−13.59	6483.94	131.0	−15.16
$150c-20f-12v$	4626.47	784.9	−84.02	4581.01	802.4	−84.18	4567.16	541.4	−84.23	4493.58	554.2	−84.48

G_1 and that G_4 is better than G_3. Furthermore, the test was applied to G_2 versus G_4. The results showed, with a significance level of less than 0.0001, that G_4 is better than G_2. These results mean that the 2nd constructive with the 2nd local search is better than the 1st constructive with the 2nd local search.

6. Modeling drone-truck delivery problems using an extension of PMS

An issue related to applying the PMS to solve PDDRP, as we showed in Section 3, is the definition of appropriate distances when

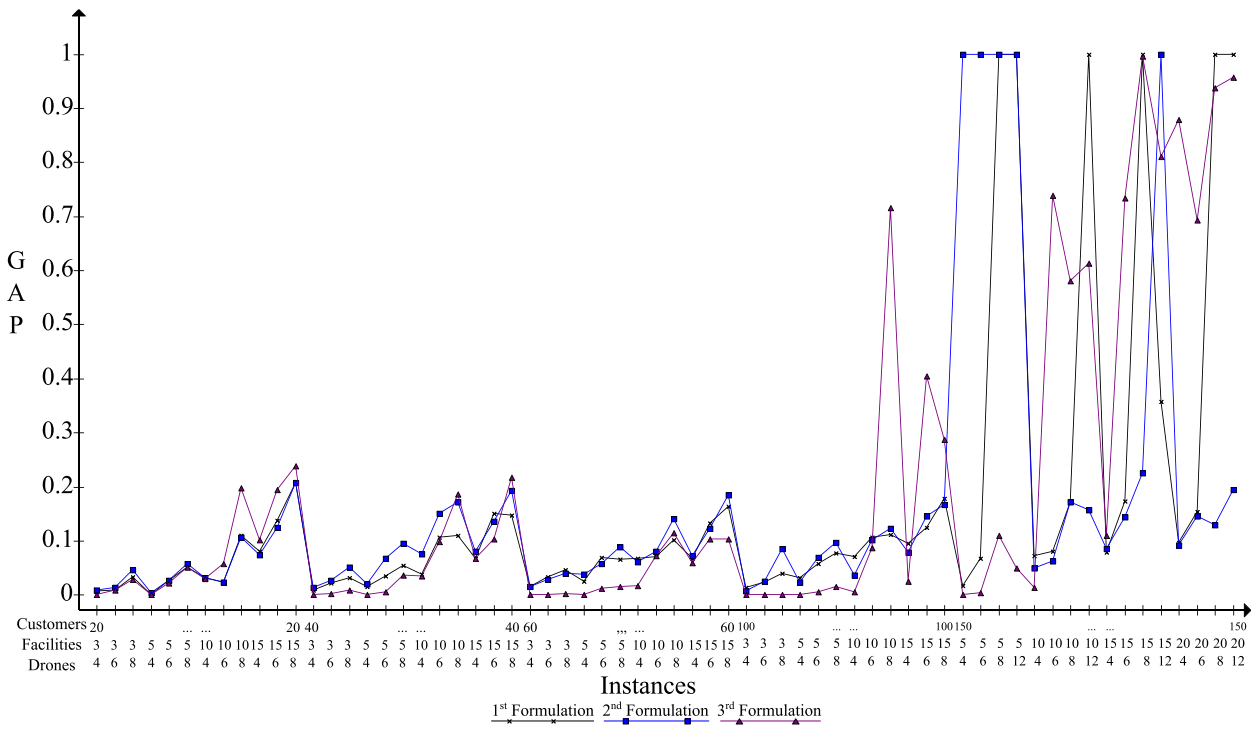


Fig. 7. A GAP comparison between the results of the three exact formulations for instances I_1 .

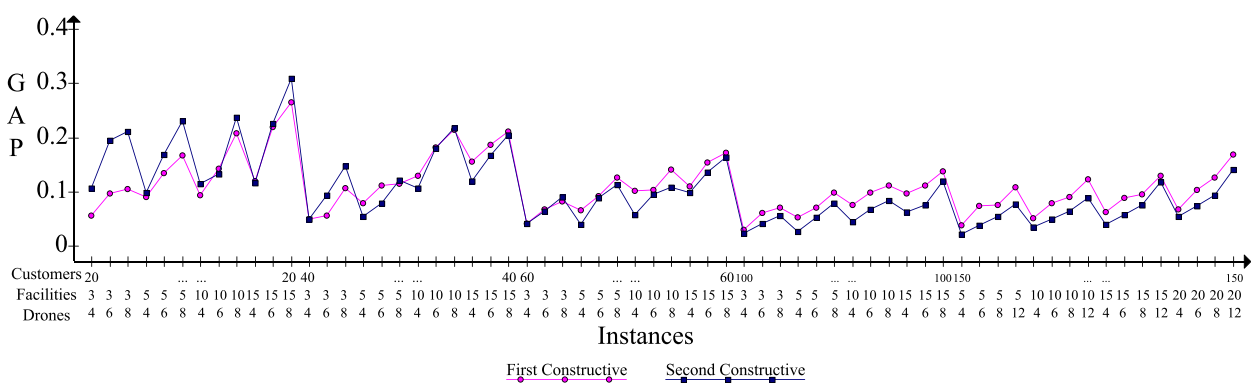


Fig. 8. A GAP comparison between the two constructive procedures for Instances I_1.

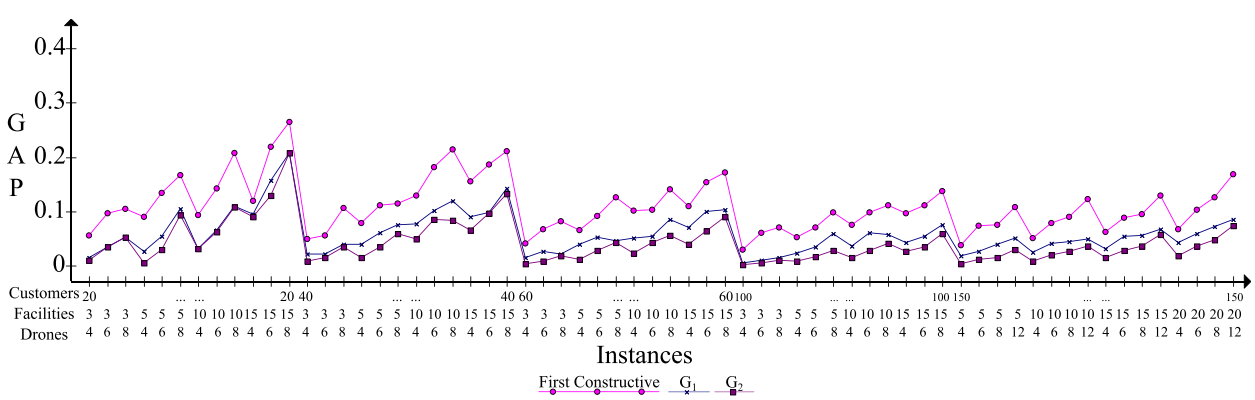


Fig. 9. A GAP comparison between first constructive procedure, and the first and second local search for Instances I_1.

Table 6
BKS for Groups 1 and 2 of Instances.

Instance	Group 1 of Instances				Group 2 of Instances			
	BKS (*units*)	LB (*units*)	GAP (%)	Best Method	BKS (*units*)	LB (*units*)	GAP (%)	Best Method
$20c-3f-4v$	3602.06	3601.70	0.01	F_3	3170.77	3170.46	0.01	F_3
$20c-3f-6v$	2469.20	2447.18	0.89	F_1 - F_3	2199.01	2172.98	1.18	F_2 - F_3
$20c-3f-8v$	1933.46	1878.16	2.86	F_3	1738.43	1691.44	2.70	F_1 - F_2 - F_3
$20c-5f-4v$	3154.77	3154.45	0.01	F_1 - F_3	2852.86	2852.58	0.01	F_3
$20c-5f-6v$	2205.20	2158.43	2.12	F_3	2044.96	1973.81	3.48	F_1 - F_2 - F_3
$20c-5f-8v$	1774.85	1684.66	5.08	F_1 - F_3	1642.03	1510.57	8.01	F_1
$20c-10f-4v$	2904.57	2816.39	3.04	F_3	2453.35	2373.49	3.26	F_3
$20c-10f-6v$	2022.67	1975.77	2.32	F_1 - F_2	1799.08	1678.40	6.71	F_1
$20c-10f-8v$	1702.29	1521.33	10.63	F_2	1559.56	1344.47	13.79	F_1 - F_2 - G_1 - G_2 - G_3 - G_4
$20c-15f-4v$	2474.44	2291.48	7.39	F_2	2284.98	2087.10	8.66	F_1 - F_2 - F_3 - G_3 - G_4
$20c-15f-6v$	1820.25	1595.21	12.36	F_2	1668.28	1533.05	8.11	F_1 - F_2
$20c-15f-8v$	1598.47	1268.31	20.65	F_1 - F_2 - G_1 - G_2 - G_4	1467.48	1205.30	17.87	G_1 - G_2
$40c-3f-4v$	7141.50	7138.26	0.05	F_3	6031.85	6029.13	0.05	F_3
$40c-3f-6v$	4789.50	4777.59	0.25	F_3	4067.19	4054.87	0.30	F_3
$40c-3f-8v$	3636.22	3602.56	0.93	F_3	3112.83	3076.16	1.18	F_3
$40c-5f-4v$	5903.75	5897.19	0.11	F_3	5244.44	5243.92	0.01	F_3
$40c-5f-6v$	4022.93	4003.65	0.48	F_3	3561.43	3544.69	0.47	F_3
$40c-5f-8v$	3135.53	3023.59	3.57	F_3	2838.74	2661.08	6.26	F_3
$40c-10f-4v$	4479.40	4323.32	3.48	F_3	4206.74	4087.63	2.83	F_3
$40c-10f-6v$	3177.12	2929.94	7.78	G_3	2966.99	2783.19	6.19	F_3
$40c-10f-8v$	2453.30	2249.50	8.31	G_2	2397.53	2143.61	10.59	G_2
$40c-15f-4v$	4416.57	4174.17	5.49	G_4	3908.56	3681.56	5.81	G_4
$40c-15f-6v$	3049.42	2850.29	6.53	G_4	2763.54	2522.22	8.73	G_4
$40c-15f-8v$	2458.40	2193.81	10.76	G_3 - G_4	2220.32	1949.34	12.20	G_4
$60c-3f-4v$	10456.90	10454.41	0.02	F_3	9279.35	9278.72	0.01	F_3
$60c-3f-6v$	6988.51	6981.76	0.10	F_3	6216.73	6205.80	0.18	F_3
$60c-3f-8v$	5265.49	5251.53	0.27	F_3	4695.51	4667.88	0.59	F_3
$60c-5f-4v$	8578.80	8574.33	0.05	F_3	7839.75	7829.85	0.13	F_3
$60c-5f-6v$	5804.46	5737.00	1.16	F_3	5294.96	5272.38	0.43	F_3
$60c-5f-8v$	4408.93	4343.33	1.49	F_3	4077.99	3943.84	3.29	F_3
$60c-10f-4v$	7834.18	7701.44	1.69	F_3	6267.04	6095.94	2.73	F_3
$60c-10f-6v$	5351.32	5179.30	3.21	G_4	4368.30	4110.43	5.90	G_4
$60c-10f-8v$	4112.86	3929.03	4.47	G_4	3399.29	3129.54	7.94	G_2
$60c-15f-4v$	6459.99	6260.42	3.09	G_4	5667.25	5445.88	3.91	G_4
$60c-15f-6v$	4485.36	4232.02	5.65	G_4	3937.34	3686.24	6.38	G_4
$60c-15f-8v$	3462.28	3227.54	6.78	G_4	3103.63	2806.61	9.57	G_4
$100c-3f-4v$	17003.79	17001.81	0.01	F_3	15803.28	15801.70	0.01	F_3
$100c-3f-6v$	11347.36	11343.04	0.04	F_3	10540.92	10538.61	0.02	F_3
$100c-3f-8v$	8527.38	8516.71	0.13	F_3	7923.13	7907.62	0.20	F_3
$100c-5f-4v$	13949.17	13947.86	0.01	F_3	13461.75	13456.63	0.04	F_3
$100c-5f-6v$	9361.41	9302.72	0.63	F_3	9028.80	8959.51	0.77	F_3
$100c-5f-8v$	7110.97	7000.68	1.55	F_3	6820.16	6732.22	1.29	F_3
$100c-10f-4v$	12589.35	12509.00	0.64	F_3	10612.49	10445.61	1.57	G_4
$100c-10f-6v$	8567.39	8363.89	2.38	G_4	7203.57	6991.85	2.94	G_4
$100c-10f-8v$	6511.00	6301.30	3.22	G_4	5512.97	5270.47	4.40	G_4
$100c-15f-4v$	10460.03	10292.05	1.61	G_4	9328.37	9148.43	1.93	G_4
$100c-15f-6v$	7121.67	6889.39	3.26	G_4	6389.74	6132.03	4.03	G_4
$100c-15f-8v$	5468.18	5203.55	4.84	G_4	4871.71	4631.45	4.93	G_4
$150c-5f-4v$	20864.86	20862.82	0.01	F_3	19709.47	19707.58	0.01	F_3
$150c-5f-6v$	13972.99	13912.59	0.43	F_3	13201.04	13123.90	0.58	F_3
$150c-5f-8v$	10580.44	10455.67	1.18	G_4	9993.77	9846.51	1.47	G_4
$150c-5f-12v$	7177.20	7015.72	2.25	G_4	6770.42	6577.56	2.85	G_4
$150c-10f-4v$	19257.72	19124.40	0.69	G_4	15206.97	15033.93	1.14	G_4
$150c-10f-6v$	12931.37	12769.15	1.25	G_4	10239.60	10032.09	2.03	G_4
$150c-10f-8v$	9796.44	9592.92	2.08	G_4	7762.05	7539.67	2.86	G_4
$150c-10f-12v$	6623.55	6439.83	2.77	G_4	5312.67	5062.06	4.72	G_4
$150c-15f-4v$	16040.52	15847.57	1.20	G_4	13619.41	13439.00	1.32	G_4
$150c-15f-6v$	10808.37	10588.57	2.03	G_4	9209.42	8982.09	2.47	G_4
$150c-15f-8v$	8186.08	7976.07	2.57	G_4	6997.78	6761.22	3.38	G_4
$150c-15f-12v$	5615.34	5373.18	4.31	G_4	4805.19	4549.83	5.31	G_4
$150c-20f-4v$	14711.77	14485.30	1.54	G_4	12575.35	12401.91	1.38	G_4
$150c-20f-6v$	9949.94	9680.30	2.71	G_4	8546.74	8298.41	2.91	G_4
$150c-20f-8v$	7582.64	7294.73	3.80	G_4	6483.94	6252.61	3.57	G_4
$150c-20f-12v$	5209.33	4919.95	5.55	G_4	4493.58	4216.13	6.17	G_4

* F_1 = First Formulation, F_2 = Second Formulation, F_3 = Third Formulation, G_1 = First Constructive and First Local Search, G_2 = First Constructive and Second Local Search, G_3 = Second Constructive and First Local Search, G_4 = Second Constructive and Second Local Search.

a drone flight between different types of nodes in the network. We will now extend this idea, in a more general formulation, to modeling two well-known problems, the Parallel Drone Scheduling Traveling Salesman Problem (PDSTSP) (Murray & Chu, 2015), and the Parallel Drone Scheduling Traveling Salesman Problem with Drop-Pickup (PDSTSP^{+DP}) (Ham, 2018).

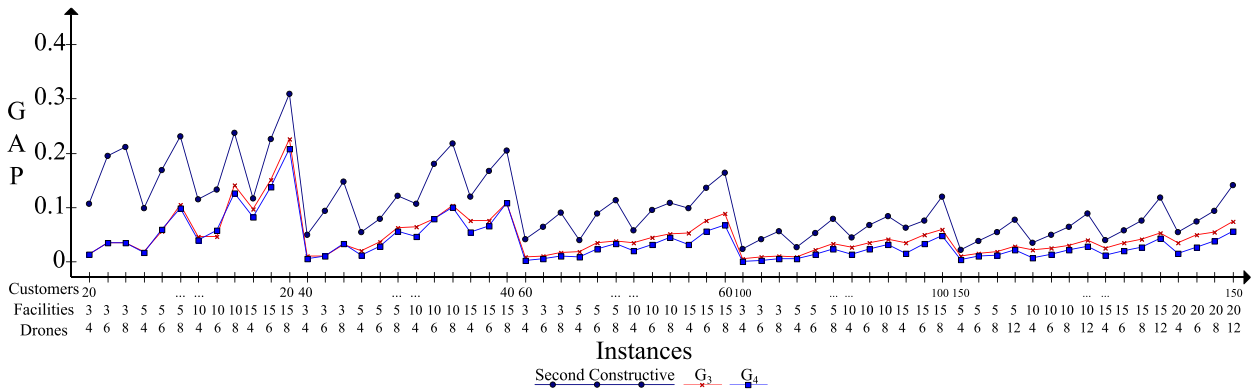


Fig. 10. A GAP comparison between second constructive procedure, and the first and second local search for Instances I_1.

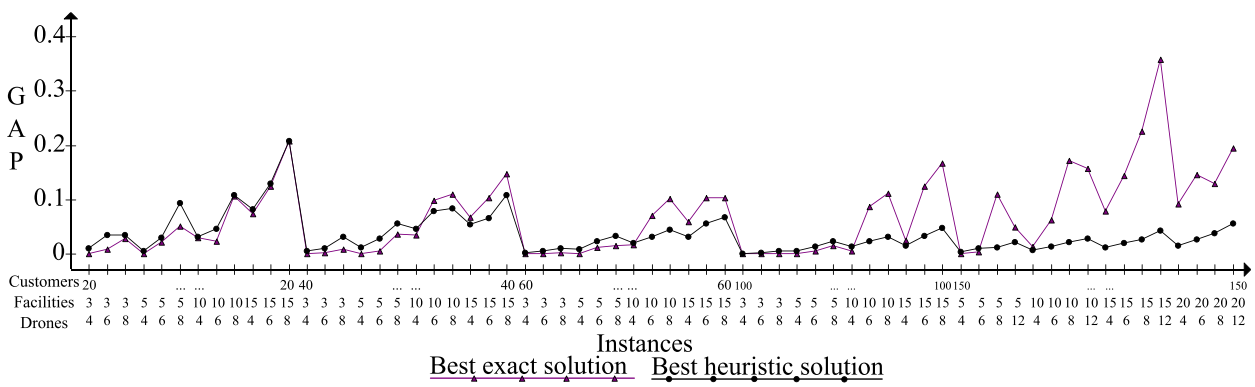


Fig. 11. A GAP comparison between the best exact solution and the best GRASP solution for Instances I_1.

Table 7
Wilcoxon signed ranks test results.

Comparison	R^+	R^-	p-value
G_1 versus G_2	8242	14	6.54×10^{-22}
G_3 versus G_4	8084	172	2.41×10^{-20}
G_2 versus G_4	6794	1462	7.45×10^{-10}

6.1. PDSTSP^{+DP} (Ham, 2018)

In the general version of this problem is defined a set of depots $\dot{E}$ from where the vehicles begin and finish their operations, and a set of customers $\dot{C} = \dot{D} \cup \dot{P}$. $\dot{D}$ is the subset of drop or delivery nodes, and $\dot{P}$ is the subset of pickup nodes. Also the notation $\dot{N} = \dot{E} \cup \dot{C}$ is used. Then, the network can be defined as $\dot{G} = (\dot{N}, \dot{A})$, where $\dot{A} = \{(i,j), i \in \dot{N}, j \in \dot{N}, i \neq j\}$. The set of vehicles is defined as $\dot{V} = \dot{T} \cup \dot{U}$; that is, $\dot{T}$ and $\dot{U}$ are the sets of trucks and drones, respectively. However, all vehicles of one type are identical. Since different types of vehicles could not use the same physical path between each pair of nodes (or because of another cost parameter incompatibilities), the original PMS distance parameters $\hat{t}_{ij}$ must be replaced for a vehicle-dependent distance parameters $\hat{t}_{ijv}$.

Both vehicles, trucks and drones, start and end its operations at a defined depot $\dot{e}_v \in \dot{E}, v \in V$. A truck travels following a road network with distances $\dot{t}_{ij}, (i,j) \in \dot{A}$ and describes a Hamiltonian cycle visiting a subset of the customer nodes. A drone uses Euclidian distances $\ddot{t}_{ij}, (i,j) \in \dot{A}$. In Ham (2018), it is commented that after a drone completes a drop, it can fly directly to another customer for pickup instead of returning to a depot and starting a new trip. Thus, this enhanced drone operation allows the possibility of reverse logistics. In the PDSTSP^{+DP}, customers must be served during a certain time window $[a_i, b_i], i \in \dot{C}$.

Fig. 12 shows how the drones' operation could be represented as a Hamiltonian cycle. On these, are illustrated two depot nodes, $\dot{e}_1$ and $\dot{e}_2$, two delivery nodes $\dot{d}_1$ and $\dot{d}_2$, and two pickup nodes $\dot{p}_1$ and $\dot{p}_2$. Fig. 12(a) indicates that a drone which starts in a determined depot, must have to fly directly to the first customer of its tour. This customer could be either a pickup or delivery node. Similarly occurs in the travel from a delivery node to a pickup node, situation exemplified in 12(b). Fig. 12(c) illustrates that a drone traveling from a delivery node to another delivery node (black line) has to use the depot that induces the minimum distance as an intermediate node (red or green line). This condition also applies in pickup-to-delivery and pickup-to-pickup travels, situations displayed in Figs. 12(d) and 12(e), respectively. This intermediate depot node is used to pick the parcels needed in travels that finish in a delivery node and to store the parcels collected in travels that begin in a pickup node. Further, the definition of the problem does not allow the use of the parcels collected into a pickup to serve a delivery node directly. Fig. 12(f) states that the last travel of the drone must be to its beginning depot.

Formally, Eqs. (37)–(44) relate PDSTSP^{+DP} with extended PMS parameters.

$$\hat{t}_{0jv} = \dot{t}_{\dot{e}_v,j}, j \in \dot{C}, v \in \dot{T} \tag{37}$$

$$\hat{t}_{ijv} = \dot{t}_{ij}, i \in \dot{C}, j \in \dot{C} : i \neq j, v \in \dot{T} \tag{38}$$

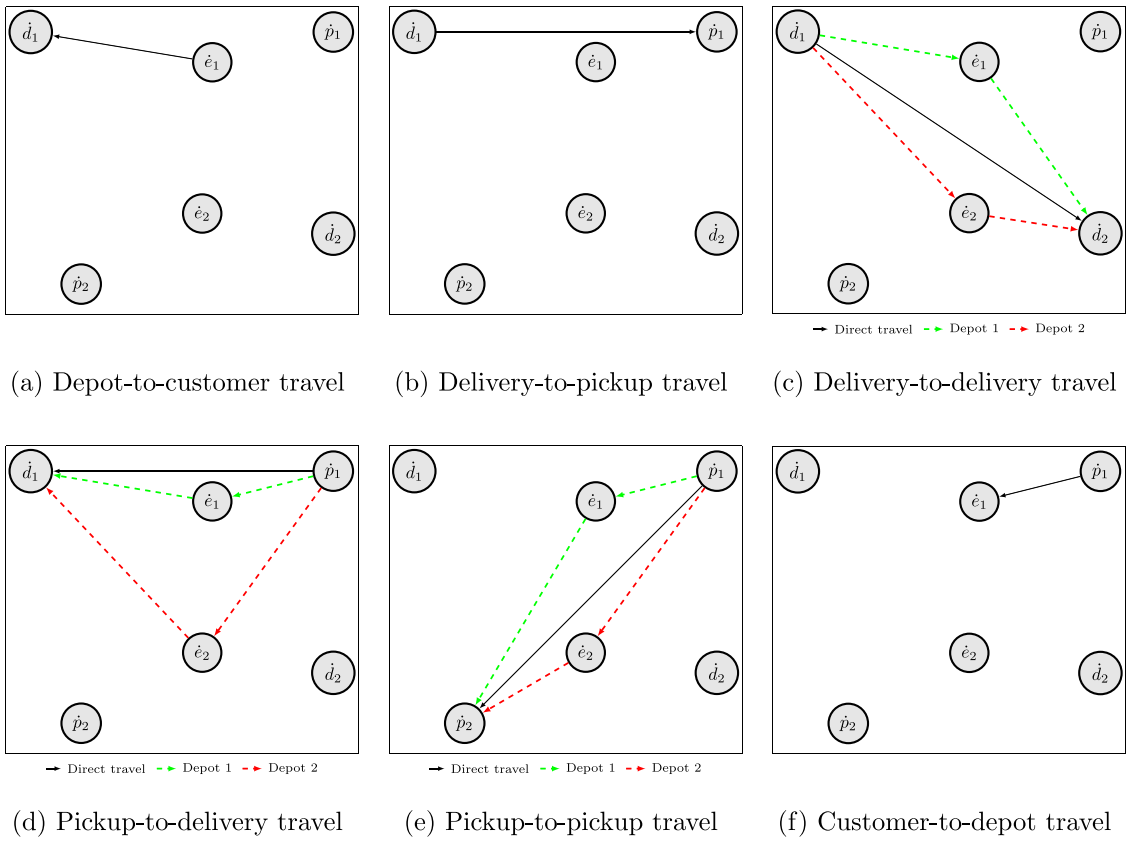
$$\hat{t}_{i,|C|+1,v} = \dot{t}_{i,\dot{e}_v}, i \in \dot{C}, v \in \dot{T} \tag{39}$$

$$\hat{t}_{0jv} = \ddot{t}_{\dot{e}_v,j}, j \in \dot{C}, v \in \dot{U} \tag{40}$$

$$\hat{t}_{ijv} = \ddot{t}_{ij}, i \in \dot{D}, j \in \dot{P}, v \in \dot{U} \tag{41}$$

$$\hat{t}_{ijv} = \min\{\ddot{t}_{i\dot{e}} + \ddot{t}_{\dot{e}j} : \dot{e} \in \dot{E}\}, i \in \dot{D}, j \in \dot{D}, v \in \dot{U} \tag{42}$$

$$\hat{t}_{ijv} = \min\{\ddot{t}_{i\dot{e}} + \ddot{t}_{\dot{e}j} : \dot{e} \in \dot{E}\}, i \in \dot{P}, j \in \dot{C}, v \in \dot{U} \tag{43}$$



(a) Depot-to-customer travel (b) Delivery-to-pickup travel (c) Delivery-to-delivery travel

(d) Pickup-to-delivery travel (e) Pickup-to-pickup travel (f) Customer-to-depot travel

Fig. 12. Example to illustrate travel distances $\hat{t}_{ijv}$.

$$\hat{t}_{i,|C|+1,v} = \ddot{t}_{i,\dot{e}_v}, i \in \dot{C}, j \in \dot{D}, v \in \dot{U} \tag{44}$$

6.2. PDSTSP (Murray & Chu, 2015)

In the PDSTSP, a depot, a set of customers, and a set of vehicles conformed by trucks and drones are defined. A truck starts in the depot and describes a Hamiltonian cycle visiting a subset of the customer nodes. A drone could visit several customers, returning to the depot after each delivery. It is well-known PDSTSP is a special case of PDSTSP^{+DP}. In fact, a PDSTSP^{+DP} instance in which $|\dot{E}| = 1$, $\dot{P} = \emptyset$ and $[a_i, b_i] = [0, \infty], i \in \dot{C}$ represents a PDSTSP instance. Therefore, this problem could also be solved using the PMS extended formulation, transforming the parameters of the distance as shown in Section 6.1.

6.3. The extended PMS model

Since PDSTSP and PDSTSP^{+DP} consider two types of vehicles, the variables $\hat{x}_{ijv}$, which initially are associated with the drones' path, change their meaning to the vehicles' path. Also, in the original model, the set V must be replaced by $\dot{V}$.

The extended PMS can be formulated using (13)–(16) and (18)–(22), replacing (17) by (17′). Also, is needed to add the constraints (45), which indicates that every customer must be served within its time window.

$$D_{jv} \geq D_{iv} - M + (\hat{t}_{ijv} + M)\hat{x}_{ijv}, (i, j) \in \hat{A}, v \in \dot{V} \tag{17'}$$

$$a_i \leq D_{iv} \leq b_i, i \in C, v \in \dot{V} \tag{45}$$

Note that if $[a_i, b_i] = [0, \infty], i \in \dot{C}$, as is in the PDSTSP, also the m-TSP formulation can be extended to solve the problem. In the context of truck-drone collaboration for delivery, the m-TSP has been used to formulate a routing problem (Kitjacharoenchai et al., 2019). It is allowed autonomous drones to fly from delivery trucks, make deliveries, and fly to any available delivery truck nearby. They proposed a MIP formulation that minimizes the makespan of both trucks and drones returned at the depot after completing the deliveries. Additionally, they presented insertion heuristics to solve large-size instances and conducted experiments to study the performance of the strategy when compared to other configurations of the vehicles.

7. Conclusions and future work

We formally define The Pickup to Delivery Drone Routing Problem (PDDRP). Its computational complexity has been determined as an NP-Hard problem, and exact and heuristic approaches have been proposed and implemented for a set of tested instances. The exact methods come from MILP formulations; one of them, the First Formulation, extends the m-TSP. The Second Formulation is an extension of a parallel machines scheduling problem. The Third Formulation has explicitly represented all the conditions of the problem. One contribution of the paper is the MILP formulations. First, Second and Third Formulations allow to find lower bounds for all instances, and hence GAPs were calculated through these bounds and the solutions provided by GRASP approaches. The First and Second Formulations got GAPs significantly smaller than those obtained by the Third Formulation in some instances, contributing to obtaining reasonable lower bounds. Moreover, the GAPs obtained in this form were small.

GRASP approaches significantly contributed to finding good solutions for the test instances. By using two variants of the greedy randomized procedure combined with two local search strategies, four GRASP variants were implemented. For all the larger instances with the greatest size, G_4 got the best-known solutions to achieve small GAPs, even for instances with 150 customers.

Table A.8
Data of facilities' locations.

Node	Instance 1		Instance 2		Region	Used Facilities				
	X-axis	Y-axis	X-axis	Y-axis		20f	15f	10f	5f	3f
f_1	587.18	429.90	510.06	494.13	R_1	x	x	x	x	x
f_2	420.71	354.31	622.13	371.90	R_1	x	x	x	x	x
f_3	508.67	449.65	409.67	365.63	R_1	x	x	x	x	x
f_4	406.86	538.27	520.62	629.99	R_1	x	x	x	x	
f_5	615.87	492.91	560.78	430.89	R_1	x	x	x		
f_6	409.22	365.82	362.41	550.78	R_1	x	x	x		
f_7	483.02	513.66	372.49	576.76	R_1	x	x	x		
f_8	484.30	578.57	388.96	500.09	R_1	x	x			
f_9	394.85	538.60	457.97	409.95	R_1	x	x			
f_{10}	467.80	603.39	427.90	456.39	R_1	x	x			
f_{11}	395.56	565.44	388.08	620.03	R_1	x	x			
f_{12}	526.34	418.63	569.47	361.67	R_1	x				
f_{13}	501.10	594.13	455.31	452.83	R_1	x				
f_{14}	580.95	636.93	443.58	510.93	R_1	x				
f_{15}	730.05	750.99	637.80	238.86	R_2	x	x	x	x	
f_{16}	724.40	597.47	203.44	309.74	R_2	x	x	x		
f_{17}	761.10	257.56	295.84	750.65	R_2	x	x			
f_{18}	508.69	242.02	783.28	366.99	R_2	x				
f_{19}	569.66	840.49	820.10	270.04	R_3	x	x	x		
f_{20}	420.82	84.63	601.28	85.28	R_3	x				

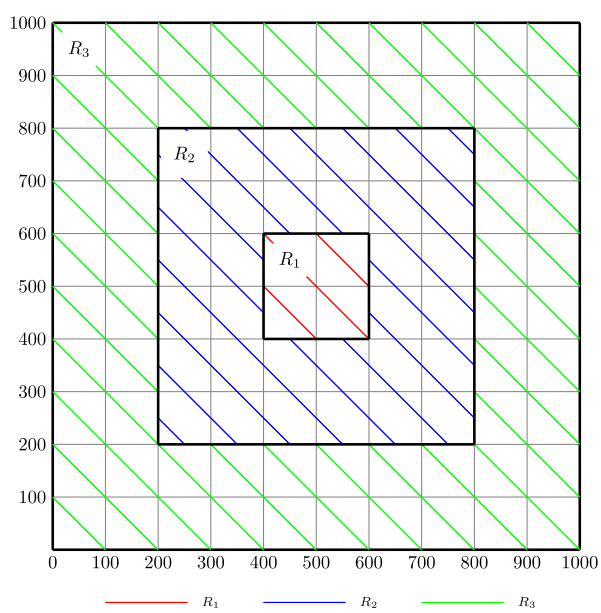


Fig. A.13. Square region and sub-region distribution.

The extended PMS model presented in the previous section is interesting because allows to formulate complex truck-drone collaboration for delivery problems. New transformations for the extended PMS model could be explored in the future. Also, we think there is room for designing new algorithms to improve the PDDRP's BKSs. One interesting avenue for future work is to consider an extension of the problem where incomplete inventory in the facilities exists. In this context, the customer order would be a list with several products, and the key assumption is that not necessarily all the products are available in one facility. Then, a drone could visit more than one facility to complete the customer's order in such a case. The extension needs extra effort in modeling the drone's route since several pickup drone operations could be required to satisfy the customer order.

CRediT authorship contribution statement

Javier Gómez-Lagos: Conceptualization, Methodology, Investigation, Formal analysis, Validation, Software, Writing – reviewing. **Benjamín Rojas-Espinoza:** Methodology, Software, Data curation, Validation, Writing – reviewing. **Alfredo Candia-Véjar:** Conceptualization, Methodology, Writing – original draft, Writing – reviewing, Supervision.

Data availability

Data are provided in Appendix

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Appendix. Test instances

As is presented in Section 5.1, two sets of instances are generated within a 2-D space separated into three concentric squares. These squares form three different regions named R_1, R_2 and R_3 whose are distributed as is shown in Fig. A.13.

Table A.8 shows the coordinates, i.e. X-axis and Y-axis, of all the facilities generated with respect to the 2-D space, for the main groups of instances 1 and 2. Additionally, is defined the region to which each facility corresponds. Since the followed instance generation strategy is accumulative, is also described which of all facilities are considered in each variation of the instance's sizes $|F| \in \{3, 5, 10, 15, 20\}$. Analogously, in the context of customers is showed in Tables A.9 and A.10 for the instance's sizes $|C| \in \{20, 40, 60, 100, 150\}$. As is mentioned above, the depot is always positioned in (500, 500). The Eq. (A.1) specifies how to calculate the Euclidean distance t_{ij} between a pair of nodes (i, j), considering x_i and y_i as the values in X-axis and Y-axis, respectively, for each node i. For practical purposes, t_{ij} is rounding to its second decimal.

$$t_{ij} = \sqrt{(x_i - x_j)^2 + (y_i - y_j)^2} \tag{A.1}$$

Table A.9
Data of customers' locations (part 1).

Node	Instance 1		Instance 2		Region	Used Customers				
	X-axis	Y-axis	X-axis	Y-axis		150c	100c	60c	40c	20c
c_1	461.46	492.39	510.91	520.89	R_1	x	x	x	x	x
c_2	394.42	483.63	406.33	568.73	R_1	x	x	x	x	x
c_3	532.56	475.73	428.54	352.08	R_1	x	x	x	x	
c_4	625.32	360.28	508.95	648.93	R_1	x	x	x	x	
c_5	391.65	352.27	373.72	451.85	R_1	x	x	x		
c_6	377.14	427.27	409.80	524.74	R_1	x	x	x		
c_7	360.63	426.11	493.44	442.05	R_1	x	x			
c_8	504.20	466.74	407.73	493.72	R_1	x	x			
c_9	580.46	522.06	621.96	410.11	R_1	x	x			
c_{10}	472.41	491.81	627.20	492.02	R_1	x	x			
c_{11}	482.91	632.42	465.37	514.46	R_1	x				
c_{12}	585.46	449.40	587.14	421.87	R_1	x				
c_{13}	574.73	405.11	468.25	426.17	R_1	x				
c_{14}	384.27	580.29	385.45	547.63	R_1	x				
c_{15}	435.67	436.18	530.54	494.43	R_1	x				
c_{16}	400.88	309.86	286.11	712.25	R_2	x	x	x	x	x
c_{17}	781.42	480.44	217.75	678.22	R_2	x	x	x	x	x
c_{18}	614.49	286.51	471.70	253.91	R_2	x	x	x	x	x
c_{19}	748.01	225.69	445.66	316.48	R_2	x	x	x	x	x
c_{20}	300.47	598.91	499.73	725.32	R_2	x	x	x	x	
c_{21}	274.71	767.85	284.98	374.74	R_2	x	x	x	x	
c_{22}	208.69	219.67	264.16	215.61	R_2	x	x	x	x	
c_{23}	239.56	349.86	424.33	316.29	R_2	x	x	x	x	
c_{24}	400.60	756.27	658.69	479.11	R_2	x	x	x		
c_{25}	526.37	753.72	685.42	230.77	R_2	x	x	x		
c_{26}	517.68	672.47	229.51	353.18	R_2	x	x	x		
c_{27}	548.58	259.46	424.82	272.42	R_2	x	x	x		
c_{28}	587.16	281.51	248.22	665.79	R_2	x	x			
c_{29}	236.52	776.01	538.58	235.42	R_2	x	x			
c_{30}	651.30	516.18	794.38	746.88	R_2	x	x			
c_{31}	388.73	235.02	715.83	511.67	R_2	x	x			
c_{32}	723.16	634.18	447.64	213.42	R_2	x	x			
c_{33}	349.19	330.35	790.22	686.45	R_2	x	x			
c_{34}	779.74	285.31	251.27	613.71	R_2	x	x			
c_{35}	213.70	303.19	226.42	257.92	R_2	x	x			
c_{36}	277.65	424.57	676.05	437.95	R_2	x				
c_{37}	555.64	211.71	204.66	218.78	R_2	x				
c_{38}	689.11	224.15	460.55	702.05	R_2	x				
c_{39}	209.93	341.81	751.05	236.36	R_2	x				
c_{40}	611.61	328.35	613.65	291.32	R_2	x				
c_{41}	422.48	211.37	765.43	208.54	R_2	x				
c_{42}	451.32	674.50	516.77	762.23	R_2	x				
c_{43}	280.35	681.56	774.90	413.33	R_2	x				
c_{44}	284.36	616.66	711.35	218.03	R_2	x				
c_{45}	511.79	758.00	680.16	762.89	R_2	x				
c_{46}	39.70	249.98	878.96	760.33	R_3	x	x	x	x	x
c_{47}	902.46	704.50	691.49	192.44	R_3	x	x	x	x	x
c_{48}	66.82	962.99	91.85	424.54	R_3	x	x	x	x	x
c_{49}	845.04	53.31	218.71	940.79	R_3	x	x	x	x	x
c_{50}	698.05	912.73	911.19	281.00	R_3	x	x	x	x	x
c_{51}	92.14	574.08	806.25	895.82	R_3	x	x	x	x	x
c_{52}	964.67	99.95	850.26	88.76	R_3	x	x	x	x	x
c_{53}	873.76	143.83	198.88	884.18	R_3	x	x	x	x	x
c_{54}	14.64	231.48	271.76	14.65	R_3	x	x	x	x	x
c_{55}	913.18	372.04	387.77	128.40	R_3	x	x	x	x	x
c_{56}	538.84	942.35	860.94	533.42	R_3	x	x	x	x	x
c_{57}	915.36	425.65	859.20	884.54	R_3	x	x	x	x	x
c_{58}	869.19	514.34	875.34	526.04	R_3	x	x	x	x	x
c_{59}	327.44	973.54	187.97	232.32	R_3	x	x	x	x	x
c_{60}	795.99	93.02	930.26	322.46	R_3	x	x	x	x	
c_{61}	113.97	614.37	844.16	176.01	R_3	x	x	x	x	
c_{62}	93.39	266.06	383.62	37.16	R_3	x	x	x	x	
c_{63}	779.30	916.01	925.13	555.27	R_3	x	x	x	x	
c_{64}	886.08	114.10	476.69	7.49	R_3	x	x	x	x	
c_{65}	928.94	191.53	170.14	409.15	R_3	x	x	x	x	
c_{66}	727.96	919.00	116.13	457.51	R_3	x	x	x	x	
c_{67}	80.17	912.75	0.44	645.35	R_3	x	x	x	x	
c_{68}	657.87	70.14	179.45	899.40	R_3	x	x	x	x	
c_{69}	935.27	257.97	814.32	976.94	R_3	x	x	x	x	
c_{70}	868.08	482.24	849.45	310.96	R_3	x	x	x	x	
c_{71}	192.28	890.89	691.82	176.31	R_3	x	x	x	x	
c_{72}	811.89	371.71	127.02	397.59	R_3	x	x	x	x	
c_{73}	563.37	986.23	624.49	937.89	R_3	x	x	x	x	
c_{74}	743.74	805.57	810.96	225.31	R_3	x	x	x		
c_{75}	891.01	282.38	944.27	520.97	R_3	x	x	x		

Table A.10
Data of customers' locations (part 2).

Node	Instance 1		Instance 2		Region	Used Customers				
	X-axis	Y-axis	X-axis	Y-axis		150c	100c	60c	40c	20c
c_{76}	464.41	38.82	940.94	69.47	R_3	x	x	x		
c_{77}	876.63	503.97	186.42	933.88	R_3	x	x	x		
c_{78}	137.02	312.84	387.64	940.20	R_3	x	x	x		
c_{79}	802.27	118.10	622.48	807.21	R_3	x	x	x		
c_{80}	127.79	485.96	944.95	19.83	R_3	x	x	x		
c_{81}	889.29	648.92	612.24	109.81	R_3	x	x	x		
c_{82}	340.78	920.39	956.25	969.20	R_3	x	x	x		
c_{83}	14.08	5.14	569.63	986.85	R_3	x	x	x		
c_{84}	465.73	198.72	70.05	385.70	R_3	x	x	x		
c_{85}	10.38	910.05	600.64	48.08	R_3	x	x	x		
c_{86}	28.82	35.08	104.64	797.84	R_3	x	x	x		
c_{87}	461.46	867.80	78.38	227.83	R_3	x	x	x		
c_{88}	899.14	210.62	596.16	31.13	R_3	x	x			
c_{89}	72.44	585.00	536.21	949.92	R_3	x	x			
c_{90}	965.05	130.20	881.02	459.68	R_3	x	x			
c_{91}	558.61	995.30	218.53	45.01	R_3	x	x			
c_{92}	934.80	389.47	428.75	951.18	R_3	x	x			
c_{93}	11.77	491.66	155.23	732.65	R_3	x	x			
c_{94}	185.26	301.25	979.01	405.46	R_3	x	x			
c_{95}	940.91	286.07	582.79	54.95	R_3	x	x			
c_{96}	70.19	245.58	25.45	954.86	R_3	x	x			
c_{97}	982.35	241.91	106.52	170.94	R_3	x	x			
c_{98}	147.62	614.42	292.28	807.80	R_3	x	x			
c_{99}	456.37	145.21	296.09	864.46	R_3	x	x			
c_{100}	324.61	907.20	83.87	283.87	R_3	x	x			
c_{101}	857.29	412.37	914.64	474.47	R_3	x	x			
c_{102}	810.63	130.02	275.23	123.49	R_3	x	x			
c_{103}	297.82	22.83	991.90	303.05	R_3	x	x			
c_{104}	721.78	891.37	904.34	579.36	R_3	x	x			
c_{105}	498.98	981.48	831.23	527.28	R_3	x	x			
c_{106}	340.92	42.28	362.58	903.55	R_3	x	x			
c_{107}	517.24	127.71	186.34	958.28	R_3	x	x			
c_{108}	224.64	915.45	103.41	321.03	R_3	x	x			
c_{109}	115.28	70.66	893.10	707.73	R_3	x	x			
c_{110}	139.99	515.97	582.21	800.27	R_3	x	x			
c_{111}	370.20	104.53	507.62	186.38	R_3	x	x			
c_{112}	66.59	17.18	118.59	389.40	R_3	x	x			
c_{113}	777.03	947.82	984.93	832.29	R_3	x	x			
c_{114}	326.12	962.52	67.65	666.01	R_3	x	x			
c_{115}	596.88	879.09	937.25	89.70	R_3	x	x			
c_{116}	128.79	724.86	600.55	931.78	R_3	x				
c_{117}	87.79	276.17	852.58	801.92	R_3	x				
c_{118}	998.43	391.65	804.67	601.46	R_3	x				
c_{119}	259.51	85.58	908.33	214.81	R_3	x				
c_{120}	656.57	865.65	112.90	356.01	R_3	x				
c_{121}	873.95	829.11	108.49	734.05	R_3	x				
c_{122}	210.78	924.94	811.90	209.08	R_3	x				
c_{123}	633.86	107.64	993.72	565.26	R_3	x				
c_{124}	144.52	596.71	798.01	41.34	R_3	x				
c_{125}	880.08	729.68	705.19	899.98	R_3	x				
c_{126}	961.30	296.95	966.99	306.15	R_3	x				
c_{127}	893.22	381.28	920.75	355.12	R_3	x				
c_{128}	134.00	532.29	803.96	336.96	R_3	x				
c_{129}	823.09	29.22	590.27	108.82	R_3	x				
c_{130}	223.06	70.64	724.40	49.08	R_3	x				
c_{131}	18.46	648.36	860.40	561.81	R_3	x				
c_{132}	33.02	423.64	820.54	910.78	R_3	x				
c_{133}	114.41	973.83	147.19	83.35	R_3	x				
c_{134}	743.74	946.52	149.18	498.91	R_3	x				
c_{135}	46.38	597.79	168.62	594.58	R_3	x				
c_{136}	912.15	103.12	779.72	144.75	R_3	x				
c_{137}	928.39	315.28	140.24	456.91	R_3	x				
c_{138}	193.25	594.48	759.35	851.52	R_3	x				
c_{139}	113.01	806.88	990.59	191.27	R_3	x				
c_{140}	949.84	533.44	168.81	917.27	R_3	x				
c_{141}	993.06	774.57	982.24	718.22	R_3	x				
c_{142}	60.08	91.54	159.16	311.92	R_3	x				
c_{143}	901.78	53.49	905.32	797.97	R_3	x				
c_{144}	58.62	400.90	643.39	191.45	R_3	x				
c_{145}	228.78	197.88	581.48	882.60	R_3	x				
c_{146}	814.36	936.71	135.10	977.06	R_3	x				
c_{147}	733.46	152.24	54.82	150.61	R_3	x				
c_{148}	111.52	460.74	162.36	183.10	R_3	x				
c_{149}	551.97	27.36	912.28	390.03	R_3	x				
c_{150}	90.60	988.57	904.52	194.12	R_3	x				

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